



MASTER THESIS

Parametric Study of the Nano E-Drive Mechanism in Clamped-Clamped Beams

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Computational Materials Science at the Technische Universität Bergakademie Freiberg.

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Manikandan Valliappan

LIST OF ABBREVIATIONS AND SYMBOLS

ABBREVIATIONS

- MEMS - Microelectromechanical systems
- Bio-MEMS - Bio medical Microelectromechanical systems
- RFMEMS - Radio-frequency microelectromechanical system
- LCD - Liquid Crystal Display
- NED - Nanoscopic Electrostatic Drive
- APDL - Ansys Parametric Design Language
- MAPDL - Mechanical Ansys Parametric Design Language
- FEM – Finite element method
- RA – Regression Analysis
- DA –Dimensional Analysis
- SPL - Sound Pressure Level
- THD – Total Harmonic Distortion

SYMBOLS

- L - Length of the beam
- b - Thickness of the beam
- h - Depth of the beam
- g_0 - Electrostatic gap of the beam
- σ - Recess
- S - Arc length of S shaped beam
- α - Angle defining the span of the electrodes
- r - Radius of curvature
- p - Pressure
- E - Young's modulus
- μ - Poisons ratio
- G - Shear modulus
- ρ - Density
- m - Mass
- ϵ_0 - Relative permittivity
- ϵ_r - Vacuum permittivity

- U - Voltage
- U_{in} – Input voltage
- w - Deflection
- Q - Charge
- ω_n - Resonance frequency
- f_{EM} - Electromechanical Resonance Frequency
- C_0 - capacitor
- k_1 - Spring stiffness 1
- k_1 - Spring stiffness 2
- k_1 - Spring stiffness 3
- $[M]$ - Mass Matrix
- $[K]$ - Structural Stiffness Matrix
- $[K^{hooke}]$ - Structural stiffness matrix
- $[K^s(\sigma(u))]$ - Stress Stiffening Matrix
- $[K^{eu}]$ - Electrostatic Softening Matrix
- $[K^{ev}]$ - Coupling Matrix
- $[K^d]$ - Dielectric permittivity coefficient matrix
- $\{f\}$ - Force Vector
- $\{u\}$ - Deflection Vector
- $\{F^e\}$ - Nodal electric force vector
- $\{q\}$ - Charge vector
- β - Unknown parameters
- X_i - Independent variables
- Y_i - Dependent variable
- e_i - Error terms

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CHAPTER 1

INTRODUCTION

Microelectromechanical systems (MEMS) are small components that combine logic elements and micromechanical structures in one chip. They process mechanical and electrical information. Micro Electromechanical systems are used in various fields like silicon angular rate sensors, small power motors, piezoelectric fuel injection valves in automobile engineering, and microphones, headphones, loudspeakers, LCD projectors in communication technology.

These mechatronic chips are made up of silicon. The structures can be smaller than one micrometer. If the main features of the electromechanical system are in the micrometer range, then the system is considered a MEMS system. Robustness, long-term stability, and consistent product quality are often features of MEMS. Moreover, their advantages include small size, lower price, and low energy consumption. Recent advances in the design and modeling of microelectromechanical systems (MEMS) enable pioneering devices for medical applications (Bio-MEMS) [13] and communication systems (RF MEMS) [14].

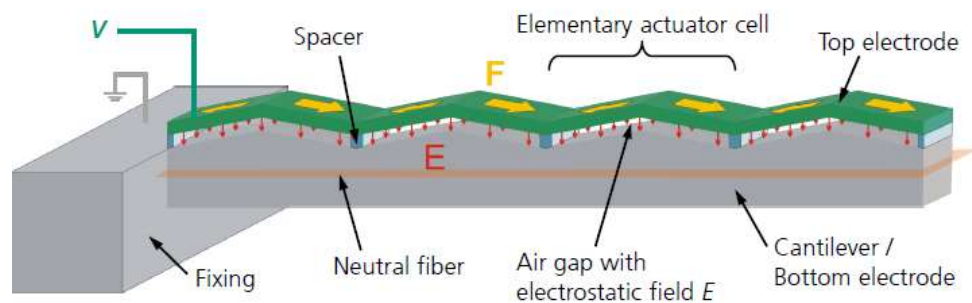


Figure 1 : Electrostatic bending actuators [11]

Nanoscopic Electrostatic Drive (NED) is an electrostatic actuation mechanism based on Microelectromechanical systems (MEMS) technology. An electrostatic actuator consists of two parallel placed electrodes and a small gap between them. When voltage is applied, an electrostatic field is generated between these electrodes, which creates an attractive force between these electrodes. These electrostatic electrodes, in turn, transform electrostatic forces into mechanical forces. Therefore, large deflections of electrostatic actuation can be achieved with small electrode separation and low voltage.

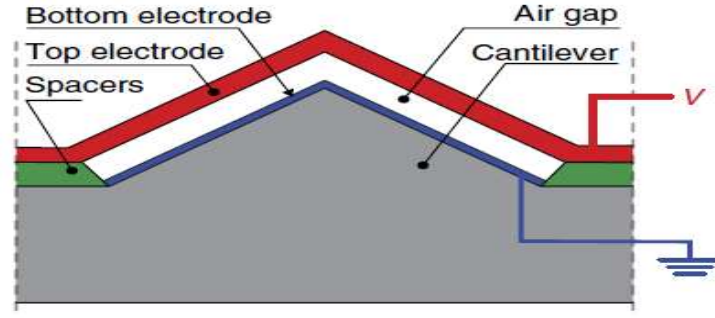


Figure 2 : Schematic image of Λ -shaped NED [3]

Figure 2 shows the electrostatically actuated Λ -shaped NED actuator. The Λ -shaped NED actuator is made up of regularly repeated electrostatic-force-based elementary actuator cells. These elementary actuator cells transform the electrostatic forces into lateral mechanical forces, which tend to bend a cantilever. The single actuator cell consists of a base with two electrodes are placed on it. These electrodes are separated by spacers and air gaps. The difference in electrostatic potential applied, generates an electrostatic force that bends the top electrode with downward displacement. On the other hand, spacers block the top electrode from large deflection. However, it produces a small strain in the top electrode. The top electrode with this surface strain deforms the entire actuator and tends to bend the cantilever. From this structure, it has been proved that the deflection of a cantilever depends on electrostatic force and very small electrode separation causes higher deflection [3].

Advantages of the Nano E-drive mechanism are easy manufacturing, low power consumption, which made them an attractive choice for many applications. When a voltage is applied to the NED beam shaped structure, it tends to bend and produces large deflections. This NED principle is used to fabricate MEMS-based micro-positioner stages [4], loudspeakers [5], peristaltic pumps [6], and other applications.

1.1 MOTIVATION

Nano electrostatic drive (NED) uses electrostatic forces to produce significant deflections in the structure. However, these actuators possess many nonlinearities from Coulomb force and stress-stiffening phenomena that cause the instability of key performance variables such as Pull-in voltage, Deflection at pull-in, Resonance frequency. Because of the complex geometry of the system, it made it very difficult to describe with an ab initio model. Therefore, the design of NED-based electrostatic actuators depends on the FEM simulation and requires optimization techniques to optimize its performances.

Though the genetic algorithm by Schuffenhauer, 2018 [7] is robust, efficient, and flexible, but it does not offer a clear idea about how the individual geometric variables influence the actuator's performance. Therefore, there is a requirement for a model that relates the main geometrical parameters with the key performance variables. Such a model could also allow finding scaling laws by which alternative fabrication processes can be assessed.

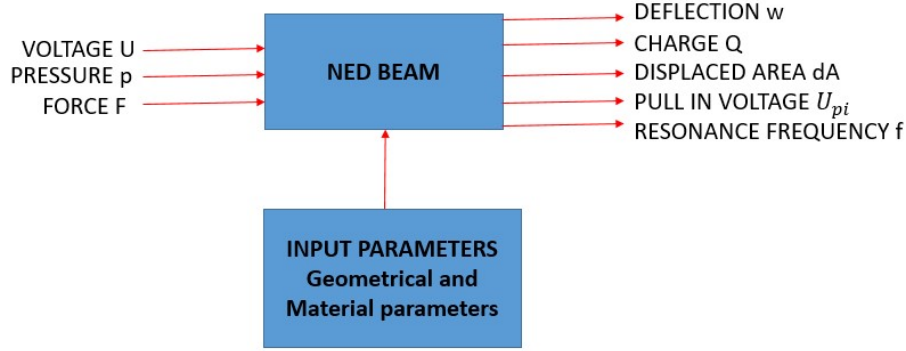


Figure 3 : Process diagram of NED beam

From this viewpoint, an empirical model of this structure can be derived by performing a parametric sweep of a finite-element model. With the outcome of such simulations, a regression can be applied to the data to find numerical relationships between the performance variables and main geometrical parameters. The performance variables depending on the dimensions of the main geometrical parameters can cause the model to be inhomogeneity and dimensionally dependent. Using dimensionless variables to find the coefficients to fit the model will be constant no matter what scales and units of measurement are selected. Dimensional analysis can be achieved by using Buckingham's Pi theorem [8]. As in every physical system, the variables can be reduced to a set of dimensionless ratios, as formulated in Buckingham's Pi theorem. Therefore, amount of variables that need to be analysed for regression is reduced.

Since the analytical solution of Coulomb-actuated prismatic beam is known [15], dimensional and regression analysis can be verified. Furthermore, it can be extended to complex structures like the S-shaped NED beam to predict the empirical relationship of performance variables.

Therefore, this thesis focuses on an extensive parametric study of the prismatic beam and S-shaped NED clamped-clamped structure using FEM simulation and analyzing the results by Buckingham's Pi theorem [8] to find causal relationships between its geometrical parameters and its electromechanical performance.

1.2 OBJECTIVES

- Developing an Ansys APDL script for the FEM simulation of a prismatic, Coulomb-actuated micro beam (clamped-clamped)
- Extending it for a parametric study.
- Comparing the results of the parametric study with the available analytic model of this structure.
- Developing an Ansys APDL script of the clamped-clamped S-shaped NED structure
- Performing a parametric study on it, including all relevant dimensions.
- Finding empirical laws of both prismatic beam and S-shaped NED beam that relates the following performance variables with the geometrical parameters:
 1. Pull-in voltage
 2. Charge
 3. Resonance frequency
 4. Effective acoustic area
 5. Electromechanical resonance frequency
 6. Displaced volume

CHAPTER 2

METHODOLOGY

In the research and development of MEMS technology, the design and simulation of Nano electrostatic drive play a vital role. In this study, the finite element analysis of the NED micro beam based on MEMS (Micro-Electro-Mechanical Systems) is modelled in detail.

To understand the NED mechanism, let us consider an example of a parallel plate capacitor suspended by a spring. Generally, asymmetric electro-static transducers are modelled as a spring capacitor system. A parallel plate capacitor with an initial gap applied with a voltage across the electrode tends to displace the moving electrode towards the fixed one. Such displacement reduces the gap between the electrodes. But it is opposed by the force of a polynomial spring. Hence, higher-order terms should be introduced to model the stress-stiffening effects. Though the spring capacitor system gives a good approximation for electrostatic transducers, parallel plate capacitor imposes few limitations. In this case, when deformation exceeds one-third of the initial gap of the parallel plate capacitor, the pull-in occurs.

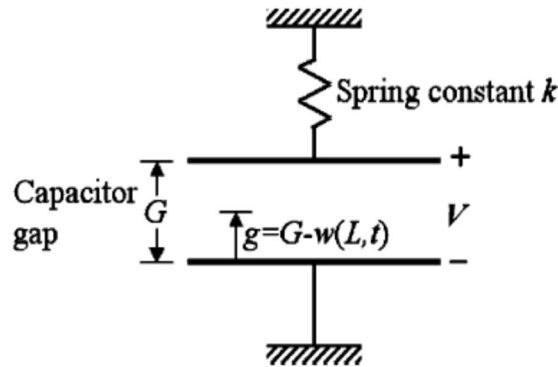


Figure 4 : Parallel-plate capacitor suspended with a spring [9]

Nano electrostatic drive shows significant deflection of an elastic structure by using electrostatic actuation. When voltage is applied to the NED beam, the electrostatic force will be created. This electrostatic force deflects the beam downwards while the elastic restoring force will restore the beam upward. This leads to the new equilibrium position and vibrates the beam. The large electrostatic force with a small electrostatic gap will lead to large deflections. The elastic restoring force is proportional to the deformation of the beam whereas the electrostatic force is proportional to the square of the deformation.

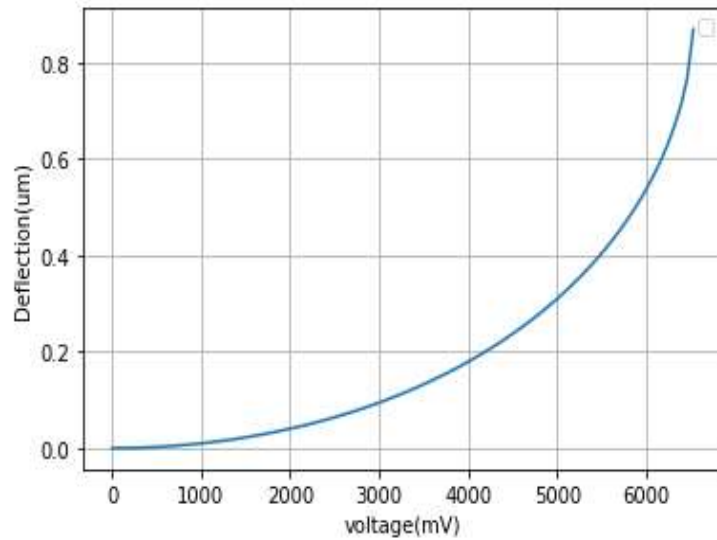


Figure 5 : Plot between voltage and deflection

The above graph shows the plot between voltage and deflection. When the applied voltage is beyond the limit, the mechanical restoring force no longer resists the opposing force and leads to the collapse of the NED beam. This phenomenon is called pull-in, and the critical voltage is called pull-in voltage. At pull-in, the beam will be in an unstable equilibrium state. The resonant frequency is the natural frequency of a system, which tends to vibrate at a higher amplitude. External loads such as pressure and force will shift their natural frequency. Many studies based on the problem of a NED beam deflected by DC electrostatic force and driven by an AC force have been investigated. Zook et al. [12] proved experimentally that increasing the driving AC voltage increases the resonance frequency.

From the FEM simulation, key performance variables such as mechanical resonance frequency, deflection, pull-in voltage, charge, displaced volume, electromechanical resonance frequency are calculated. Since all these performance variables are dimensionally dependent, it is difficult to find the empirical relationship. Hence, dimensional analysis of Buckingham's Pi theorem is applied to the performance variables to remove the dimensional homogeneity.

Buckingham's Pi theorem states that if there were n variables in a problem and these variables contain m primary dimensions (for example, M, L, T), then the equation relating all the variables will have $(n-m)$ dimensionless groups [8]. The main advantage of using dimensional analysis is the number of variables is reduced for the calculation. Therefore, it makes the regression analysis effective and dimensionally independent.

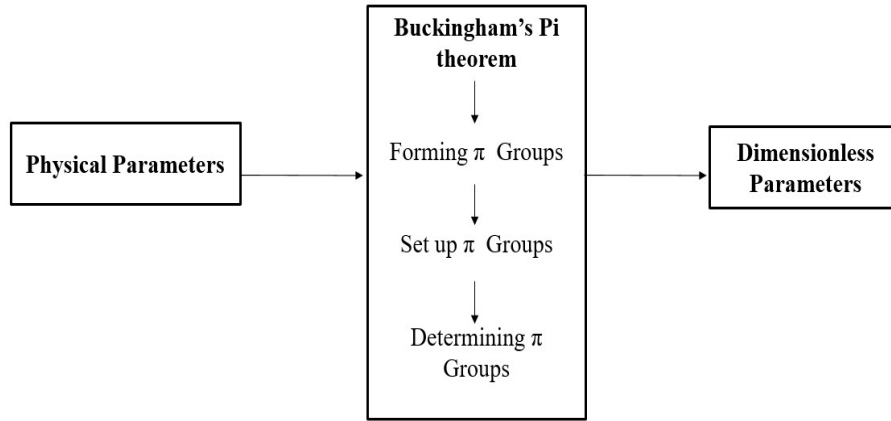


Figure 6 : Dimensional analysis

To investigate and optimize the behaviour of the prismatic, Coulomb-actuated micro beam, an FEA model has been developed using the Mechanical Ansys Parametric Design Language (MAPDL) module from ANSYS 19.0. The displacement of the beam is larger than a tenth of its thickness. Therefore, the model is solved as a non-linear model. Furthermore, key performance variables from FEM simulation are converted into dimensionless variables by using Buckingham's Pi theorem. Finally, regression analysis is applied to find the empirical relationship of key performance with geometrical parameters. Results are verified with an analytical solution and extended to the S-shaped NED beam.

A detailed description of FEM implementation and Regression analysis are described in the following sections.

2.1 FINITE ELEMENT IMPLEMENTATIONS

2.1.1. PRISMATIC BEAM

A coulomb-actuated prismatic beam is a microscopic beam in which cross section will be uniform throughout the structure.

The coulomb-actuated micro beam is shown in the figure below. It consists of two electrodes. One electrode is fixed and another electrode is free to oscillate. These electrodes are separated initially by an electrostatic gap g_0 . The fixed electrode has a curve structure with the additional parameter recess S . Free electrode has a Length L , Thickness b , and depth h . An electric potential U is applied between them.

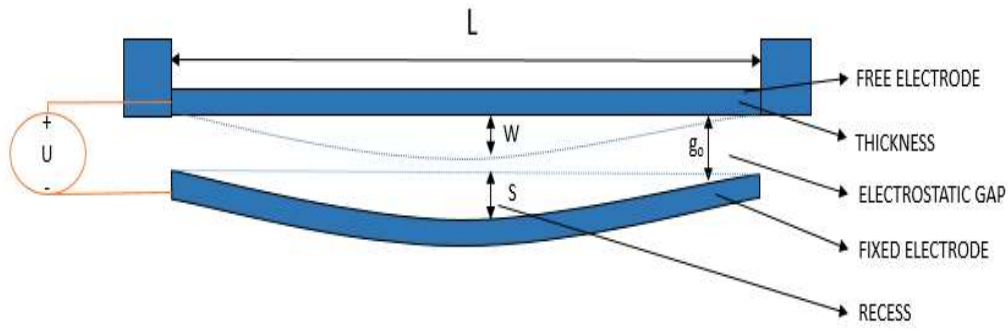


Figure 7 : Diagrammatic representation of prismatic beam

Following are the input geometrical parameters:

PARAMETER	UNIT	DESCRIPTION
g_0	$[\mu m]$	Electrostatic gap between two electrodes
s_0	$[\mu m]$	Recess describing the fixed electrode
L	$[\mu m]$	Length of the electrode
h	$[\mu m]$	Height of the electrode
b	$[\mu m]$	Thickness of the electrode

Table 1 Geometrical parameters

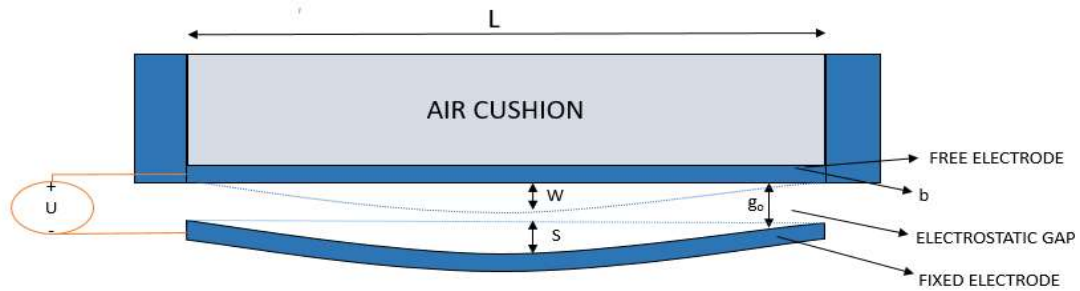


Figure 8 : Prismatic beam with air cushion

The FEM model is considered as an isotropic material. The FEM model contains two structural domains silicon and air gap. Ansys APDL doesn't have any tool for calculating the displaced area of the system. Therefore, a cushion is placed on the top of the free electrode to find the displaced volume of the system. This Air cushion has pseudo stiffness which is very small in magnitude when compared to the material of the beam.

Electrodes – silicon

Air gap – air

Air cushion – air

Material properties of silicon are as follows:

MATERIAL PARAMETERS	UNIT	VALUE
Young's modulus E	<i>MPa</i>	169e3
Poisons ratio μ	-	0.279
Shear modulus G	<i>MPa</i>	79.4e3
Density ρ	<i>kg/um³</i>	2329e-18

Table 2: Material properties of silicon

Material properties of air gap are as follows:

MATERIAL PARAMETERS	UNIT	VALUE
Young's modulus E	<i>MPa</i>	1e-7
Poisons ratio μ	-	0
Shear modulus G	<i>MPa</i>	0
Density ρ	<i>kg/um³</i>	0

Table 3: Material properties of silicon

To maintain the uniform dimension throughout the analysis, following system of units are used.

$$\mu m - \mu N - nC - kg - s - mV - \mu F$$

Meshing is an important component to obtain accurate results from a FEM model. Mesh splits the entire domain into a discrete number of elements for which the solution can be calculated. Generally, the smaller the mesh size, accurate the result. Instead of simulating with dense mesh for hours to get an accurate result, a convergence study has been carried out and added in the results and discussion chapter.

Two types of elements have been used in the simulation. Plane183 for the structural element with 8 nodes and plane223 for the electromechanical elements. Plane183 shows quadratic displacement behaviour and suits for modelling irregular meshes. Element is defined by eight nodes at beam and air cushion. Since it is a 2D simulation, plane strain is used.

In-between the beam, the air gap is considered. Since voltage is the degree of freedom for the air gap, a 3D electromechanical element is used. For the electrostatic case, the air gap has meshed with plane223 element type. To avoid mesh convergence error, Ansys recommends using the **mshmid** command for deleting the mid-side nodes in the air gap.

The ends of the prismatic beam are clamped and voltage is applied in the air gap of the beam. Additionally, pressure is as given as surface force on the top surface of the free electrode.

Governing differential Equation [2] of FEM is

$$M\ddot{u} + Ku = f \quad (1)$$

Where M is the mass matrix, K is the structural stiffness matrix, f is the force vector, and u is the deflection vector. For a simple linear matrix problem (static problem), the coefficient matrices M and K do not depend on time and deflection u.

The System becomes nonlinear when it depends on the displacement. One typical nonlinear behavior is the stress- stiffening phenomenon. Then stiffness matrix K becomes

$$K = K^{hook} + K^s(\sigma(u)) \quad (2)$$

Where K^{hook} is the structural stiffness matrix from linear Hooke's law and $K^s(\sigma(u))$ is the stress stiffening matrix depends on the deflection.

When the electrostatic fields are considered, FEM model set to have multi domain i.e. electrostatics and elastomechanics. Governing equation of coupled FEM is

$$\begin{bmatrix} M & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \ddot{u} \\ 0 \end{bmatrix} + \begin{bmatrix} K + K^{eu} & K^{et} \\ (K^{ev})^T & -K^d \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} f \\ q \end{bmatrix} \quad (3)$$

Where

- Mass matrix $[M] = \rho \int [N]^T [N] d\Omega$, N is matrix of shape functions
- Element stiffness matrix $[K] = \int [B]^T [D] [B] d\Omega$, B is strain displacement matrix
- Nodal electric force vector $\{F^e\} = - \int [B]^T \{\sigma^M\} d\Omega$, $\{\sigma^M\}$ is maxwell stress vector
- Dielectric permittivity coefficient matrix $[K^d] = \int (\nabla\{N\}^T)^T [\epsilon] (\nabla\{N\}^T) d\Omega$
- f is the force vector and q is the charge vector
- $[K^{eu}]$ and $[K^{ev}]$ are the electrostatic softening matrix and the coupling matrix calculated by derivatives of the nodal electric force $\{F^e\}$ with respect to displacement and voltage.

In Ansys APDL, three types of analysis are carried out in the FEM simulation.

1. Modal analysis – calculation of mechanical resonance frequency.
2. Static analysis – calculation of deflection, pull-in-voltage, charge and displaced area of the beam.

3. Linear perturbation analysis - calculation of electromechanical resonance frequency.

Modal analysis is a tool for knowing the vibrations of the system. Modal analysis is the dynamic analysis that gives the natural frequency at beam will resonate. Three modes of vibrations are extracted from the simulation. From which the mechanical resonance frequency of the beam is calculated. The Block Lanczos algorithm is used for modal analysis calculation, which is an algorithm for finding the null space of a matrix over a finite field.

The static analysis holds linear relation between displacement and forces applied. For a simple linear matrix problem (static problem), the coefficient matrices like the stiffness matrix do not depend on time and deflection. Here for including the stress stiffening effects, the **NLGEOM** command is used. Newton Raphson method is used for solving the nonlinear equation of the finite element discretization process. For a converged solution, more than one iteration is needed. The general algorithm of the Newton Raphson method is

On the first time step $\{u_0\} = 0$. $\{u_0\}$ is converged from the previous time step. It calculates the tangent stiffness matrix and restoring load from the from current time step. Further it calculates the $\{\Delta u_i\}$ by

$$[K_i^T]\{\Delta u_i\} = \{F^a\} - \{F_i^{nr}\} \quad (4)$$

Where

$[K_i^T]$ – Tangent stiffness matrix

i – Current iteration number

$\{F^a\}$ - Vector of applied loads

$\{F_i^{nr}\}$ – Vector of restoring loads corresponding to the element internal loads.

To obtain the next step approximation $\{u_{i+1}\}$

$$\{u_{i+1}\} = \{u_i\} + \{\Delta u_i\} \quad (5)$$

Calculation of tangent stiffness matrix, restoring load vector and next step approximation of displacement $\{u_{i+1}\}$ continuous until the convergence is obtained,

Linear perturbation analysis is the analysis to activate the perturbation procedure to determine the resonance frequency due to applied voltage. Linear perturbation analysis determines the response of the system due to perturbation loads and boundary conditions.

Parametric analysis is carried out for different set of geometrical parameters, which are as follows,

- Length = [500,600,700,800,900] μm
- Thickness = [1,2,3,4,5] μm
- Electrostatic gap = [2,2.5,3,3.5,4] μm
- Sigma = [0,0.1,0.2,0.4,0.6]
- Pressure = [-1e-6,0,1e-6] MPa

In result and discussion chapter, FEM results are discussed more in detail.

2.1.2. S-SHAPED BEAM

S-shaped NED beam is the best design for the application of micro loudspeakers. Key factors in MEMS-based micro speakers such as high sound pressure level (SPL), low total harmonic distortion (THD), and low power consumption can be achieved by using S-shaped NED actuators. Therefore, the high performance of micro speakers fulfils the customer need and the hearing aid industry.

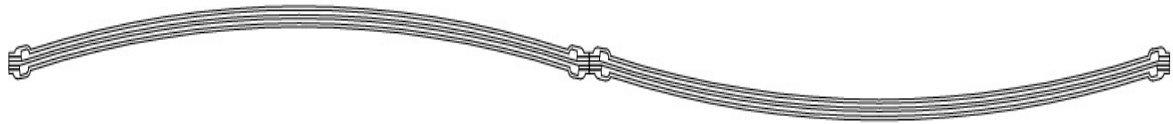


Figure 9 : S-shaped NED half beam [2]

S-shaped NED beam consists of three parallel S-shaped electrodes with different arc length S and radius of curvature r . These three electrodes are separated by electrostatic gap of g_0 . S-shaped electrodes has two sides left and right which are joined together by blockers.

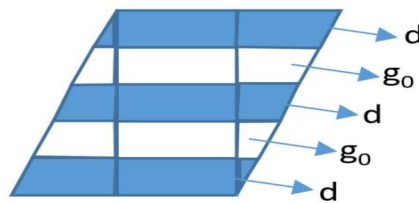


Figure 10 : Central blockers of S shaped beam

In the above figure, blue boxes are electrodes and white boxes are electrostatic gaps. To make the S-shaped beam symmetrical, parameters of both left and right-sided arcs are considered the same.

We know that,

$$\text{arc length } (S) = \theta \times r \quad (6)$$

Where θ is the angle between the two lines joining the end-points of the arc and r is the radius of curvature. Here, angle θ and arc length S are considered as the geometrical input parameter. To make the dimensions same, θ should be in radians.

Electrostatic gap, arc length of the electrode, thickness of the electrode, height of the electrode, angle defining the span of the electrodes are the geometrical input parameters of the FEM simulation. Initially, key points are created using the triangular law of vector addition. Furthermore, lines, arcs, and areas are created.

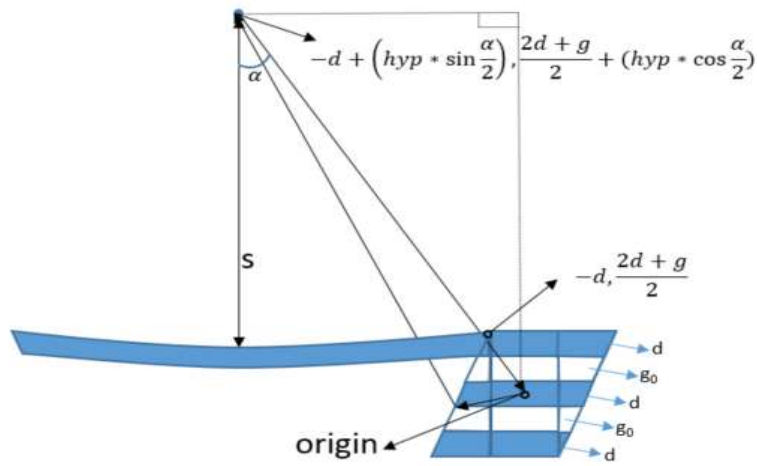


Figure 11 : Geometrical property of S shaped beam

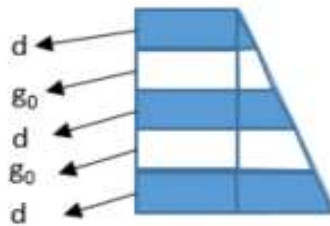


Figure 12 : End blockers of S shaped beam

Following are the input geometrical parameters:

PARAMETER	UNIT	DESCRIPTION
g_0	$[\mu m]$	Electrostatic gap between two electrodes
S	$[\mu m]$	Arc Length of the electrode
d	$[\mu m]$	Thickness of the electrode

alpha	[degree]	Angle defining the span of the electrodes
h	[μm]	Height of the electrode

Table 4 Geometrical parameters

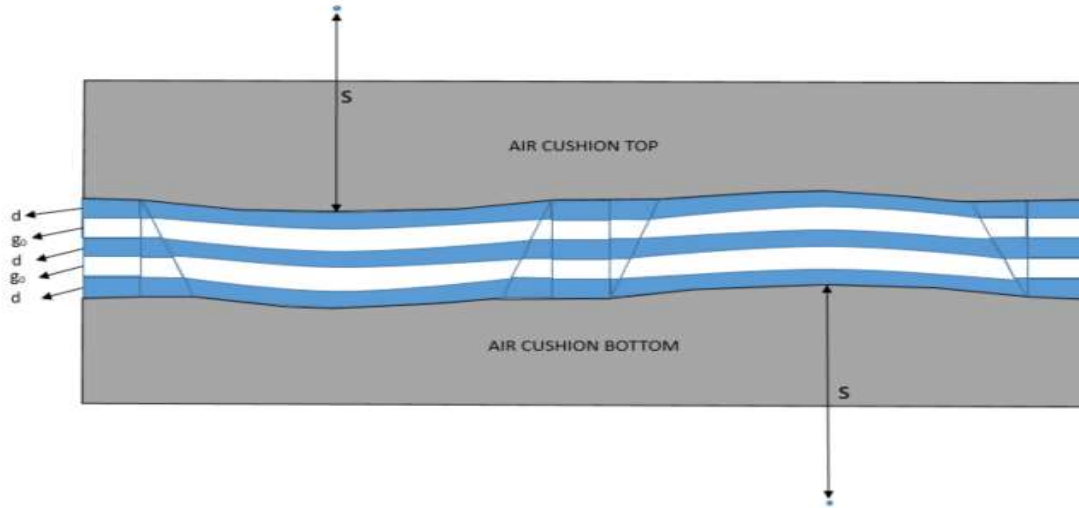


Figure 13 : Diagrammatic representation of S-shaped NED beam

Same as a prismatic beam, the model is considered as an isotropic material and has two structural domains silicon and air gap. In addition to that, a cushion is placed on the top and bottom of the electrodes to find the displaced volume of the system.

Material properties of silicon and air gap are chosen like the prismatic beam.

Same as a prismatic beam, two types of elements have been used in the simulation. Plane183 for the structural element with 8 nodes and plane223 for the electromechanical element with 8 nodes. Plane183 shows quadratic displacement behaviour and suits for modelling irregular meshes. Element is defined by eight nodes at beam and air cushion. Since it is a 2D simulation, plane strain is used.

In-between the beam, the air gap is considered. Since voltage is the degree of freedom for the air gap, a 3D electromechanical element is used. For the electrostatic case, the air gap has meshed with plane223 element type. To avoid mesh convergence error, Ansys recommends using the **mshmid** command for deleting the mid-side nodes in the air gap.

Both ends of the S-shaped beam are clamped and voltage is applied as the input in the air gap area. Modal analysis, static analysis, and linear perturbation analysis are carried out for calculation of mechanical resonance frequency, deflection, pull-in-voltage, charge, displaced area, and electromechanical resonance frequency of the beam. In the result and discussion chapter, FEM results are discussed more in detail.

Parametric analysis is carried out for different sets of geometrical parameters, which are as follows,

- Arc Length = [300,400,500,600,700] μm
- Thickness = [1,2,3,4,5] μm
- Electrostatic gap = [1,1.5,2,2.5,3] μm
- Alpha = [10,15,20,25,30] degree

2.2 DIMENSIONAL ANALYSIS OF THE ELECTROMECHANICAL SYSTEM

The performance variables depending on the dimensions of the main geometrical parameters can cause the model to be inhomogeneity and dimensionally dependent. Using dimensionless variables to find the coefficients to fit the model will be constant no matter what scales and units of measurement are selected and also main use of dimensional analysis is the number of variables is reduced. Therefore, dimensional homogeneity has to be removed before using the regression analysis. Dimensional analysis can be achieved by using Buckingham's Pi theorem [8].

Dimensional Analysis of electrostatic micro beam

$$f(\epsilon_0 \epsilon_r, E, \rho, l, R, g_0, d, U, p, w, Q, \omega_n) = 0 \quad (7)$$

Applying Buckingham's Pi theorem

$$g\left(\frac{R}{l}, \frac{d}{l}, \frac{g_0}{l}, \frac{U}{l} \sqrt{\frac{\epsilon_0 \epsilon_r}{E}}, \frac{p}{E}, \frac{w}{l}, \frac{Q}{l \sqrt{\epsilon_0 \epsilon_r E}}, \omega_n l \sqrt{\frac{\rho}{E}}\right) = 0 \quad (8)$$

Where,

ϵ_0 - Relative permittivity

ϵ_r - Vacuum permittivity

E - Young's modulus

ρ - Density

l - Length

R - Recess

g_0 - Electrostatic Gap

d - Thickness

U - Voltage

p - Pressure

w - Deflection

Q - Charge

ω_n - Resonance frequency

Proof of the dimensional analysis of analysis are added in the annexure-I.

2.3 REGRESSION ANALYSIS OF PERFORMANCE VARAIBLES

Regression analysis is an analysis used to estimate the relationship between the dependent variable and one or more independent variables by a set of statistical processes.

The regression model uses regression techniques to estimate the parameters of the model. Regression models consist of the following components,

- Unknown parameters (β)
- Independent variables (X_i)
- Dependent variable (Y_i)
- Error terms (e_i)

Most regression models propose Y_i as a function of X_i and β and e_i represents an additive error term or random statistical noise.

$$Y_i = f(X_i, \beta) + e_i, \quad i = n \text{ observations} \quad (9)$$

Regression analysis aims to estimate the function $f(X_i, \beta)$ that closely fits the data and find the parameters

2.3.1. DEFLECTION

Generally, asymmetric electro-static transducers are modeled as a spring capacitor system. A parallel plate capacitor (C_0) with an initial gap g_0 applied with a voltage U_{in} across the electrode tends to displace the moving electrode towards the fixed one. Such displacement reduces the gap between the electrodes. However, it is opposed by the force of a polynomial spring k_i . Hence, higher-order terms are introduced to model the stress-stiffening effects. The term F_{out} gives the additional loads to the system. A detailed description of analytical equations is added in the annexure 2 section. Electrical and mechanical constitutive equations [15] are,

$$q(t) = \frac{C_0}{1 - \left(\frac{w(t)}{g_0}\right)} U_{in}(t) \quad (10)$$

$$k_1 w + k_3 w^3 = \frac{1}{2} \frac{C_0}{g_0} \frac{U_{in}^2}{(g_0 - w)^2} - F_{out} \quad (11)$$

Though the spring capacitor system gives a good approximation for electrostatic transducers, parallel plate capacitor imposes few limitations. Therefore, this transducer is modeled with displacement varying capacitance with polynomial expression.

Capacitance of a prismatic beam with recess S as follows,

$$k_1 w + k_3 w^3 = \frac{1}{2} \frac{C_0}{g_0} U_{in}^2 f(w) - p S_{eff,2}^* \quad (12)$$

Where p is the pressure applied and $S_{eff,2}^*$ is the effective piston length of sensor. $f(w)$ Gives the variation of capacitance against a reference displacement.

Where,

$$f(w) = \psi^2 \left[\pi_1 + \frac{\pi_2}{\left(1 - \frac{(w - s_0)}{g_0}\right)^{\frac{1}{2}}} + \frac{\pi_3}{\left(1 - \frac{(w - s_0)}{g_0}\right)^{\frac{3}{2}}} \right] \quad (13)$$

where π being constants, $\pi_1 = \frac{\psi^2}{77}$, $\pi_2 = -\frac{\psi^2}{38}$, $\pi_3 = \frac{15\psi^2}{28}$ with $\psi = 1.5881$

$$C_0 = \frac{\epsilon_0 \epsilon_r L}{g_0} \quad (14)$$

ϵ_0 - Relative permittivity

ϵ_r - Vacuum permittivity

L - Length of the beam in μm

b - Thickness of beam in μm

w - Deflection in μm

Therefore,

$$U = \left[(k_1 w + k_3 w^3 + p S^*) \frac{2g_0}{C_0^* F\left(\frac{w}{g_0}\right)} \right]^{\frac{1}{2}} \quad (15)$$

Here U is the applied voltage from simulation. Therefore, spring constants k_1, k_3 can be calculated using the regression method. Since its nonlinear regression, Powell's method is used for least square fit. The advantage of using this method is no need for differentiating the functions. It saves time and cost from the Hessians computations of the functions. Powell method finds search directions with a small number of the error function in one direction.

Bidirectional search along each vector by golden section search method gives the accurate calculation of local minimum of the function.

These calculated spring constants k_1, k_3 are verified with the analytical solution.

$$k_1 = \frac{\lambda_0 EI}{L^3} \quad (16)$$

$$k_3 = \frac{\varepsilon_D k_1}{b^2} \quad (17)$$

Where,

E - Young's modulus in Mpa

L - Length of the beam in μm

b - Thickness of beam in μm

$$\lambda_0 = 500.604$$

$$\varepsilon_D = 0.719$$

$$I = \frac{b^3}{12}$$

In addition, effective piston length S_{eff} is calculated and verified with analytical solution

$$S_{eff} = -\frac{(k_1 w + k_3 w^3)}{p} \quad (18)$$

k_1 - Spring stiffness

w - Deflection

p - Pressure

Analytical solution for effective piston length S_{eff} is as follows

$$S_{eff} = -k_1 \frac{dw}{dp} \quad (19)$$

2.3.2. CHARGE

The charge stored by a capacitor is directly proportional to its voltage. The proportionality constant depends only on the capacitor. Moreover, it is called capacitance. Hence, polynomial regression of degree 5 with odd powers is used for the regression model to find the perfect fit.

Odd function is a function for which $f(-x) = -f(x)$. Though sign of independent variable changed but the absolute value will remain the same.

Taylor series of odd powers of degree 5 is used for regression model to find the perfect fit.

$$Q = f_1 U + f_3 U^3 + f_5 U^5 \quad (20)$$

From the above equation, f_1 is the capacitance at zero voltage. Analytical equation for capacitance at zero voltage is as follows

$$f_{1analytical} \approx \frac{\epsilon_0 L}{g_0} (1 - 0.5727\sigma + 0.3835\sigma^2) \quad (21)$$

2.3.3. DISPLACED AREA

The basic assumption of the analytical model is that only the first mode of oscillation is active, and therefore the displaced area is directly proportional to the deflection. Hence, linear regression model of $y = mx + C$ is used. Displaced area at zero voltage does not make any use for the calculation hence constant term C is forcefully made into zero.

Therefore, equation of linear regression becomes,

$$dw = m_1 * dA \quad (22)$$

2.3.4. ELECTROMECHANICAL RESONANCE FREQUENCY

For the electromechanical stiffness, we calculate the derivative of force balance equation with respect to the deflection [15].

$$\frac{d}{dw} \left(\sum F \right) = k_1 + 3k_3 w^2 - \frac{1}{2} \frac{C_0}{g_0^2} U^2 F'(z) = K_{EM} \quad (23)$$

Where,

$$F' \left(z = \frac{w}{g_0} \right) = \psi^2 \left[\frac{\pi_2}{2 \left(1 - \frac{(w - s_0)}{g_0} \right)^{\frac{3}{2}}} + \frac{3\pi_3}{2 \left(1 - \frac{(w - s_0)}{g_0} \right)^{\frac{5}{2}}} \right] \quad (24)$$

The analytical electromechanical resonance frequency is then calculated as follows:

$$f_{EM} = \frac{1}{2\pi} \sqrt{\frac{K_{EM}}{m}} \quad (25)$$

$$m = \rho b L \quad (26)$$

An approximate regression is done by proposing a second-degree polynomial. In order to describe the fall of the resonance frequency at voltages close to pull-in, a sigmoid function is artificially included.

$$f_{EM} = f_0 * \left(\frac{1}{1 + e^{\frac{U - U_{pi}}{\beta}}} \right) * (1 + (f_1 U + f_2 U^2)) \quad (27)$$

Since its nonlinear regression, Powell's method is used for finding the local minimum of the function. The advantage of using this method is no need for differentiating the functions. It saves time and cost from the Hessians computations of the functions. Powell method finds search directions with a small number of the error function in one direction. Bidirectional search along each vector by golden section search method gives the accurate calculation of local minimum of the function.

CHAPTER 3

RESULTS AND DISCUSSION

3.1. MESH CONVERGENCE

In a mesh convergence study, the FEM model is simulated with a fine and coarse mesh. And the results are compared for both cases. Therefore, it will prove that the results of the FEM model are not affected by changing the mesh size.

Number of divisions of the geometry is varied from 10 to 200 and checked for convergence.

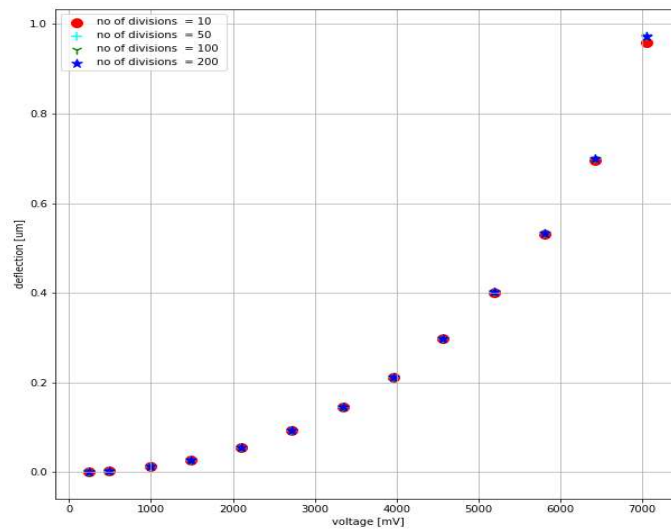


Figure 14 : Mesh convergence study - plot between voltage and deflection

From the graph, it can be understood that changing the mesh size affects the pull-in voltage and deflection. For the number of divisions is 10, pull-in occurs early. Whereas the number of divisions is 100, pull-in occurs a little late.

After the number of divisions is 100, pull-in voltage and deflection become stable. This shows that the FEM model is not affected by changing the mesh size after the number of divisions is 100.

3.2 SUBSTEP SENSITIVITY ANALYSIS

Sub steps play a vital role in the FEM simulation. It determines the divisions of load steps to run the simulation. In this analysis, a comparison of less and more number of sub-steps has been carried out and plotted below.

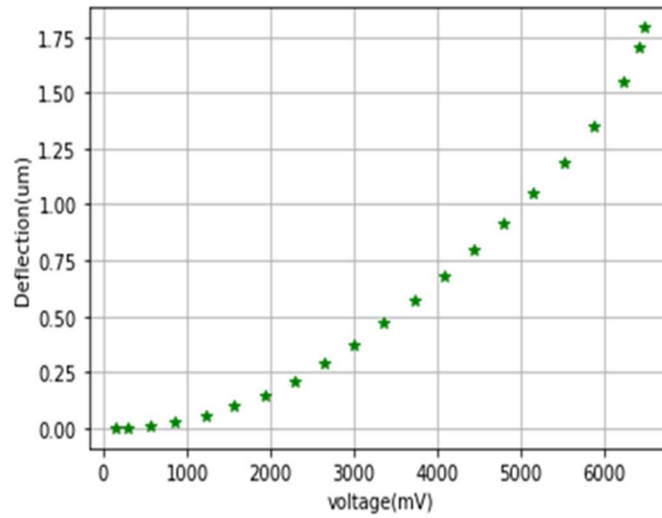


Figure 15 : Voltage vs Deflection plot with less number of sub steps

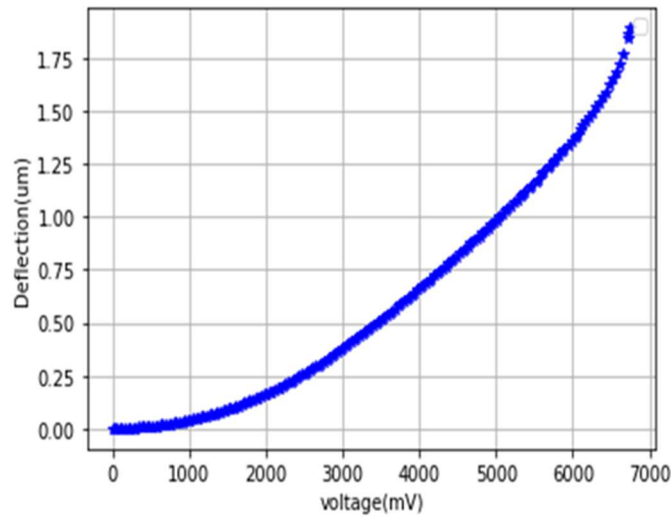


Figure 16 : Voltage vs Deflection plot with less number of sub steps

From Figures 15 and 16, it can be clearly understood that more number of sub-steps will give more information in the plot. For the regression analysis, more number of sub-steps will give the perfect fit for the regression model. Hence for this FEM simulation, the number of sub-step has been chosen to 200. All the load step values of voltage and their respective performance variables are calculated and stored in the .csv file for regression analysis.

3.3 PRESSURE SENSITIVITY ANALYSIS OF PRISMATIC BEAM

Pressure and voltage are applied at the bottom of the prismatic beam to understand the shift of pull-in voltage. Three types of pressures (negative, zero, and positive pressure) are applied. Negative pressure applied towards the prismatic beams. Therefore, it helps the voltage for deflection. Hence, the pull-in occurs early. Positive pressure is applied against the prismatic beam, which means against the voltage. Therefore, the beam requires more voltage for

deflection. Hence, the pull-in occurs a little late when compared to the negative pressure. On the other hand, zero pressure gives the proper deflection for the applied voltage. Hence, the pull-in occurs in between the positive and negative pressure.

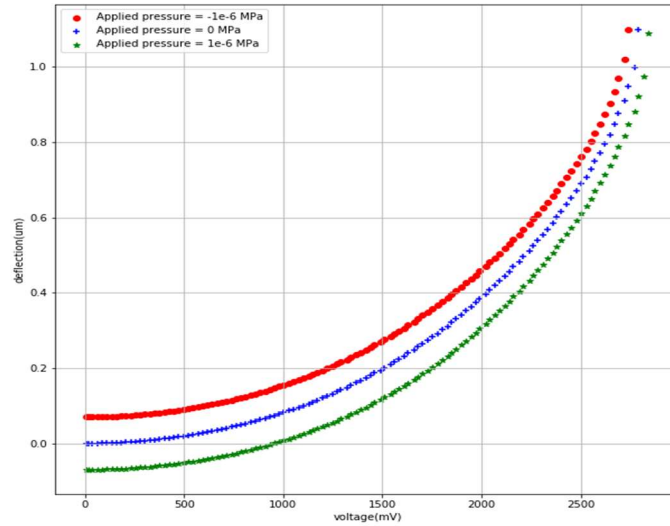


Figure 17 : Pressure analysis of prismatic beam

The plot between voltage and deflection for different pressures is shown in the graph. It shows the shift of pull-in for different pressure and voltage. For low pressure of $-1\text{e-}6$ Mpa, pull-in occurs early. For high-pressure $1\text{e-}6$ Mpa, pull-in occurs late because of pressure applied against the voltage.

3.4 PRISMATIC BEAM – COMPARATIVE STUDY

The local analysis is the analysis that gives information about how the performance variables depend on the input voltage.

Though the script is extended for the different geometrical parameters, only two sets of the parameters are taken into account to show the results.

1. Prismatic beam without recess
2. Prismatic beam with recess

3.4.1 CASE 1: Prismatic beam without recess

Prismatic beam without recess will give flat shape for the fixed electrode.

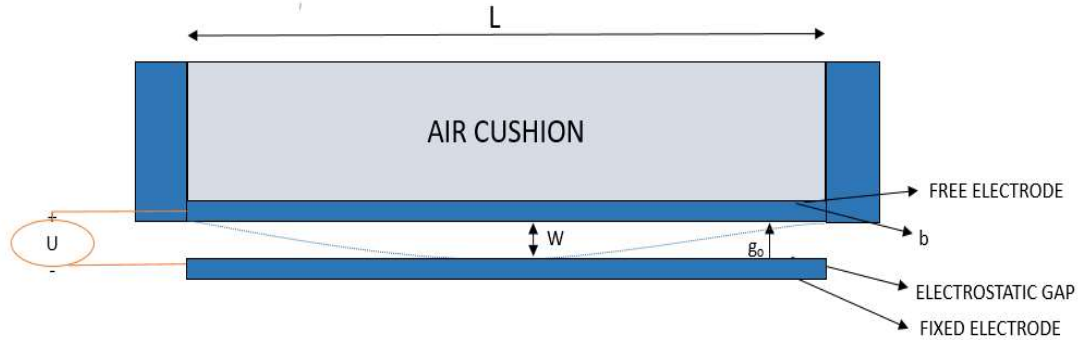


Figure 18 : Prismatic beam without recess

Parameters considered are

Length = $800 \mu m$
Thickness = $1 \mu m$
Electrostatic Gap = $2 \mu m$
Recess = 0
Pressure = $-1e-6, 0, 1e-6 Mpa$

When voltage is applied to the moving electrode, it tends to bend towards the fixed electrode. Here additionally negative, zero, and positive pressure are applied. Negative pressure applied towards the prismatic beams. Therefore, it helps the voltage for deflection. Hence, the pull-in occurs early. Positive pressure is applied against the prismatic beam, which means against the voltage. Therefore, the beam requires more voltage for deflection. Hence, the pull-in occurs a little late when compared to the negative pressure. On the other hand, zero pressure gives the deflection for the applied voltage. Hence, the pull-in occurs in between the positive and negative pressure.

As mentioned in the Finite element implementations section, the above parameters are simulated in the Ansys APDL module and results are shown below.

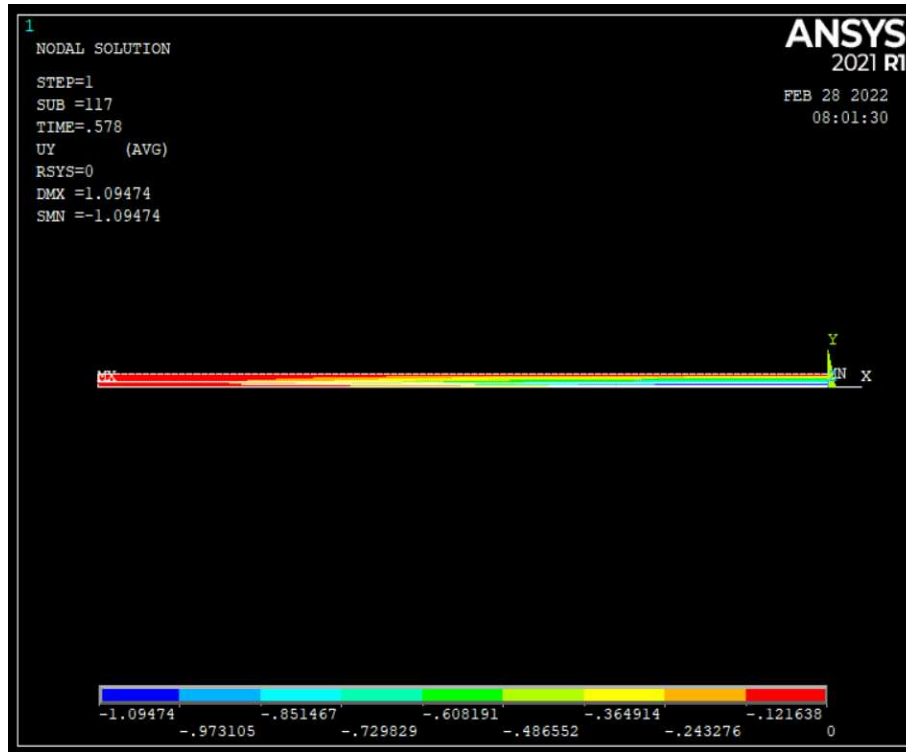


Figure 19 : FEM results of prismatic beam without recess at zero pressure

From the above FEM results, it can be clearly understood that the moving electrode displaced towards the fixed electrode in the y-direction. Additionally, the FEM script extended to the parametric study. All performance variables of each parametric set are calculated and saved in the .csv file for post-processing.

Post processing:

1. Voltage vs Deflection

When voltage is applied to the moving electrode, it tends to bend towards the fixed electrode. An additional three types of pressure are also given as input for the simulation. Max deflection at the midpoint of the prismatic beam in each sub-step voltage is calculated in the FEM simulation and stored in the .csv file. Since the FEM analysis was carried out for different parameters, the .csv file generated will have numerous data. A python file is created with pandas data frame library to access the data for the particular parameter set from the .csv file. This regression aims to predict the equation that gives deflection for each voltage. A separate routine has been created in python with Powell's method for the regression analysis. As per the Regression model discussed in the methodology section is calculated for Voltage Vs Deflection and plotted below.

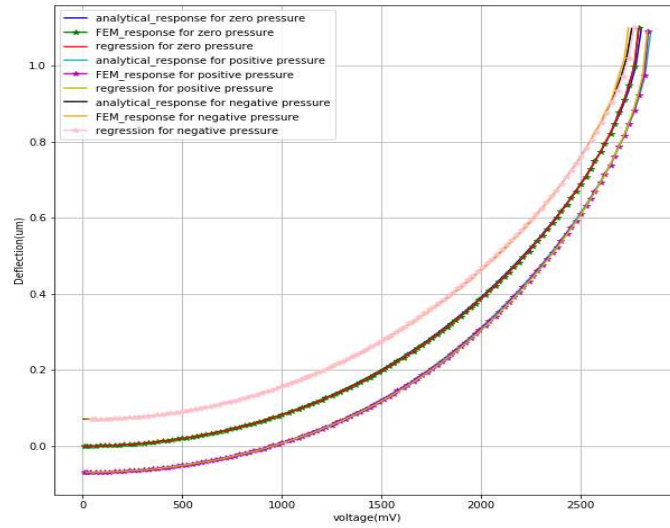


Figure 20 : Voltage vs Deflection plot for prismatic beam without recess

Analytical, simulated and regression for deflection and voltage at different pressure are compared and plotted in the graph. From the graph, it can be seen that the regression model applied is perfectly fitted with the analytical and simulation model of the prismatic beam. This shows the proposed regression model gives the best fit for the prismatic beam. To understand more about the regression model, graph between voltage and deflection is shown below for zero pressure.

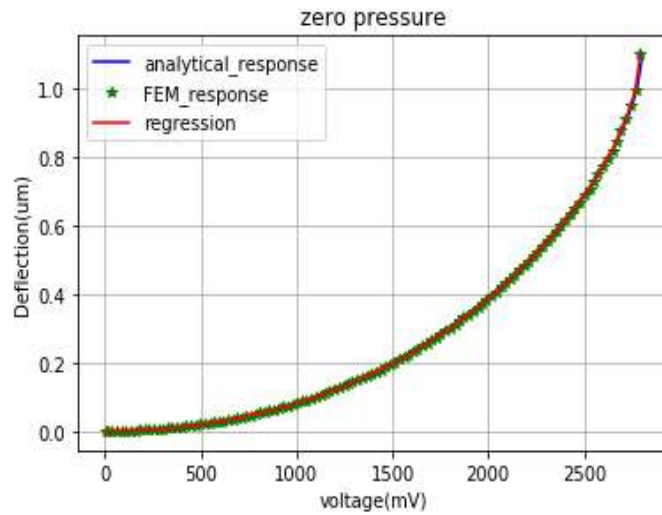


Figure 21 : Voltage vs Deflection plot at zero pressure for prismatic beam without recess

To know the fitness of regression model, errors are calculated and shown below,

Description	units	Analytical model	Regression model	Error in percentage
Spring stiffness k_1	$\mu\text{N}/\mu\text{m}$	0.01514	0.01494	0.01 %
Spring stiffness k_3	$\mu\text{N}/\mu\text{m}$	0.01038	0.010742	0.03 %
Effective piston length S_{eff}	$\mu\text{N}/\mu\text{m}/\text{MPa}$	1046.489	1042.842	0.34 %

Table 5 Calculation of error for the regression model of spring stiffness and effective piston length

Errors calculated are less than 1% .From the tabulation, it can be understood that the proposed regression model gives the perfect fit for the voltage vs deflection of the prismatic beam. The equation used in the regression gives the exact deflection for the applied voltage.

2. Deflection vs Displaced Area

To find the displaced volume from the FEM simulation an extra air cushion is considered on the top of the moving electrode and simulated. In addition, the volume before and after the loads such as voltage and pressure are calculated and subtracted to find the displaced volume of the beam. The displaced area is calculated by dividing displaced volume by height.

The basic assumption of the analytical model is that only the first mode of oscillation is active, and therefore the displaced area is directly proportional to the deflection. Therefore, linear regression model of $y = mx + C$ is used. The displaced area at zero voltage does not make any use for the calculation hence the constant term C is forcefully made into zero.

Therefore, equation of linear regression becomes,

$$dw = m_1 * dA \quad (28)$$

To check the regression model, following analytical equation used [15].

$$\frac{dA}{dw} = L * \chi_0 \quad (29)$$

where $\chi_0 = 0.523$

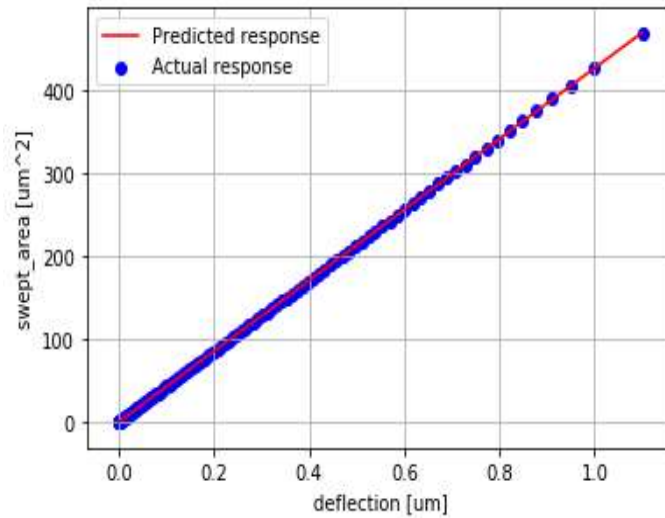


Figure 22 : Deflection vs Displaced Area for prismatic beam without recess

The above graph shows the plot between the regression model and analytical model of the displaced area and deflection. From the graph, it can be observed that the regression solution perfectly fits with the analytical solution.

Equation of regression is as follows,

$$dw = 426.7861 * dA \quad (30)$$

Description	units	Analytical model	Regression model	Error In percentage
Displaced area	μm	418.4	426.7861	2.0 %

Table 6 Calculation of error for the regression model of displaced area

Root mean square error for other set of parameters between the displaced area of regression and analytical are as follows:

Description	units	Root mean square error
Displaced area	μm	0.21%

Table 7 Root mean square of displaced area

From the tabulation, it can be understood that the proposed regression model gives the perfect fit for the displaced area vs deflection of the prismatic beam. By this regression model, the displaced area can be calculated from the deflection of the beam.

3. Voltage vs Charge

The charge of the prismatic beam in each sub-step voltage is calculated in the FEM simulation and stored in the .csv file. As per the Regression model discussed in the methodology section is calculated for Voltage Vs Charge and plotted below.

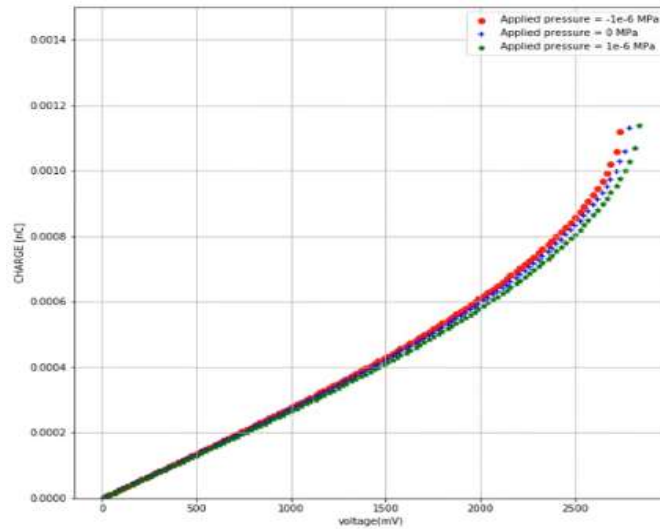


Figure 23 : Voltage vs Charge plot for prismatic beam without recess

The graph shows the plot between voltage and charge at different pressure. From the graph, it can be understood that pressure changes the charge in the beam according to the corresponding voltage.

Odd function is a function for which $f(-x) = -f(x)$. Though sign of independent variable changed but the absolute value will remain the same.

Taylor series of odd powers of degree 5 is used for regression model to find the perfect fit.

$$Q = f_1 U + f_3 U^3 + f_5 U^5 \quad (31)$$

From the above equation, f_1 is the capacitance at zero voltage. In addition, analytical equation for capacitance at zero voltage is as follows

$$f_{1_{analytical}} \approx \frac{\epsilon_0 L}{g_0} (1 - 0.5727\sigma + 0.3835\sigma^2) \quad (32)$$

Description	units	Analytical	calculated	Error in percentage
capacitance C_1	$\mu F / \mu m$	3.54167E-09	3.54911E-09	0.20 %

Table 8 Calculation of error for the regression model of Capacitance C_1

Error in the capacitance at zero voltage is 0.2%. This shows chosen regression model perfectly fits with the analytical solution. Additionally, the root mean square for other set of parameters is calculated and shown below.

Description	units	Root mean square error
Charge q	mV	1.96 %

Table 9 Root mean square of displaced area

From the tabulation, it can be understood that proposed regression model for charge gives the perfect fit for charge and voltage.

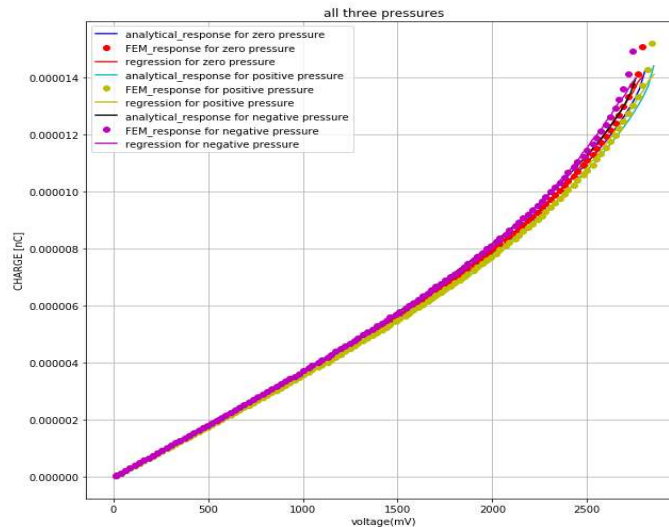


Figure 24 : Voltage vs Charge plot for prismatic beam without recess

The graph shows the plot between voltage and charge at different pressure of simulation, analytical and regression. From the graph, it can understood fitness of the chosen regression model is good.

4. Voltage vs Electro Mechanical Resonance Frequency

From linear perturbation analysis, the Electromechanical Resonance Frequency of the prismatic beam in each sub-step voltage is calculated in the FEM simulation and stored in the

.csv file. As per the Regression model discussed in the Methodology section is calculated for Voltage and Electro-Mechanical Resonance Frequency and plotted below.

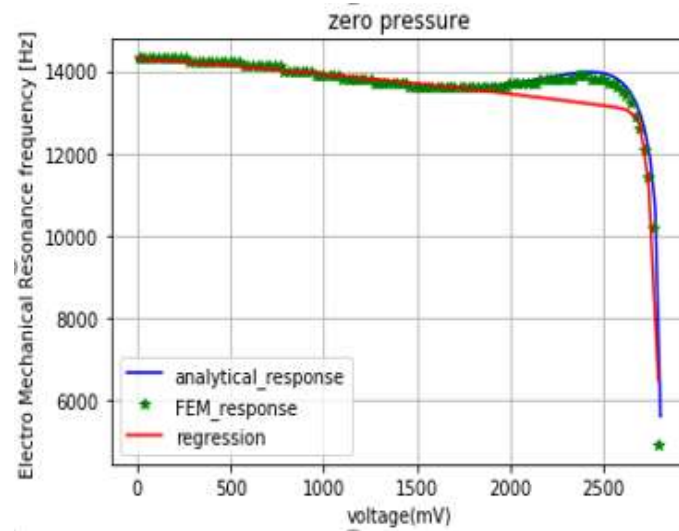


Figure 25 : Electromechanical resonance frequency vs Charge plot without recess

The graph shows the plot between voltage and Electromechanical resonance frequency at zero pressure of simulation, analytical, and regression. A deviation can be noticed in the regression model at the pull-in-region because the function fails to have the exact form of the analytical curve. An approximate function is created with the Taylor series and the sigmoid function. From an application point of view, the system cannot operate at the pull-in region. Therefore, it becomes less area of interest. Weighted regression is used in this analysis. 70 percent of data has been chosen for the regression analysis to reduce the complexity of the analysis with less order equation.

By using a higher-order equation with 100 percent of data, the regression model can be fitted with the analytical solution. However, it gives more complexity to the calculation of electromechanical resonance frequency at a particular voltage.

Description	units	Analytical	calculated	Error in percentage
Resonance frequency f_0	Hz	14259.029	14300	0.28 %

Table 10 Calculation of error for the regression model of Resonance frequency f_0

Error in the capacitance at zero voltage is 0.2. This shows the fitness of the chosen regression model.

Description	units	Root mean square error	RMSE (90% data)
Electro mechanical resonance frequency	Hz	3.68161	1.86 %

Table 11 Root mean square of electromechanical resonance frequency

From the tabulation, it can be understood that the proposed regression model for electromechanical resonance frequency gives the perfect fit for electromechanical resonance frequency and voltage. Compared to 100 percent data, the root-mean-square of 90 percent data gives the error of 1.85. A system will not operate at the pull-in voltage because it collapses the entire system. Hence, this regression model can be used for the calculation of electromechanical resonance frequency until 70% pull-in voltage.

3.4.2 CASE 2: Prismatic beam with recess

Recess in the geometrical parameter of Prismatic beam gives the curved structure for the fixed electrode.

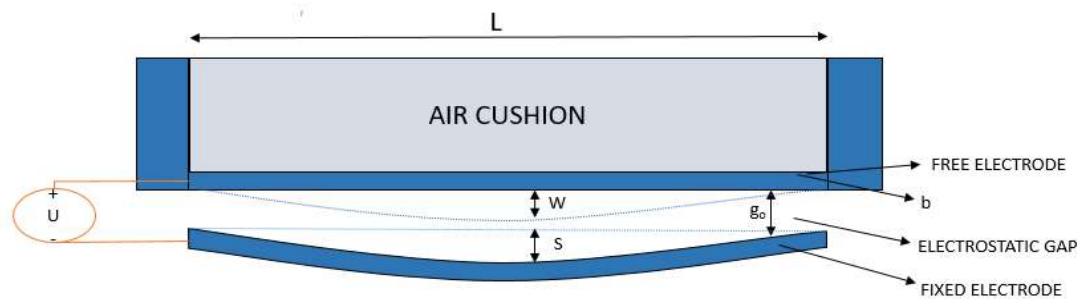


Figure 26 : Prismatic beam without recess

Parameters considered are

Length = $800 \mu m$
Thickness = $1 \mu m$
Electrostatic Gap = $2 \mu m$
Recess = Electrostatic Gap*0.6
Pressure = $-1e-6, 0, 1e-6 Mpa$

As mentioned in the Finite element implementations, above parameters are simulated in Ansys APDL module and results are shown below.

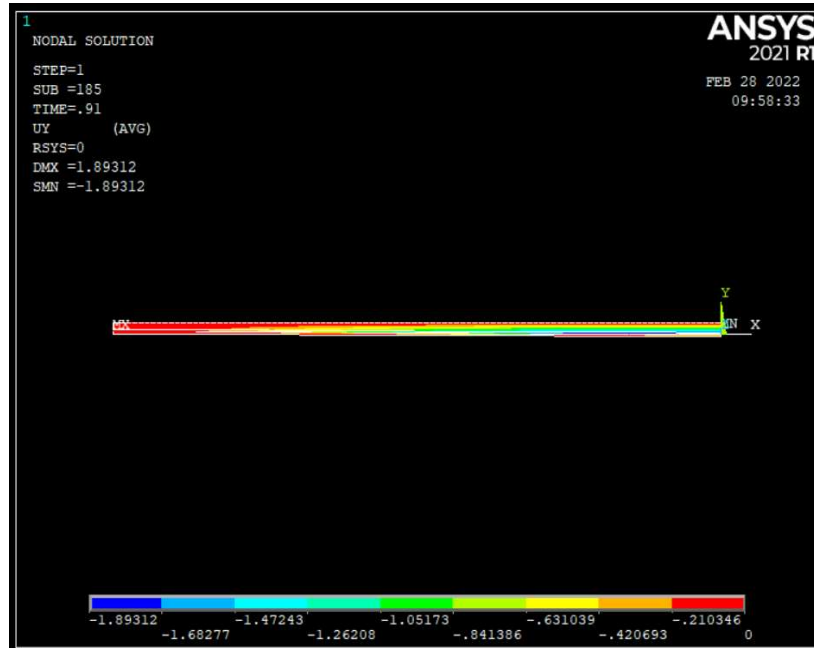


Figure 27 : FEM results of prismatic beam with recess

From the above results, it can be clearly understood that the moving electrode displaced towards the fixed curved electrode in the y-direction. All performance variables are calculated and saved in the CSV file for post-processing.

Post processing:

1. Voltage vs Deflection

By the same calculation from the previous study, the deflection of the beam is calculated from the FEM simulation and stored in a .csv file.

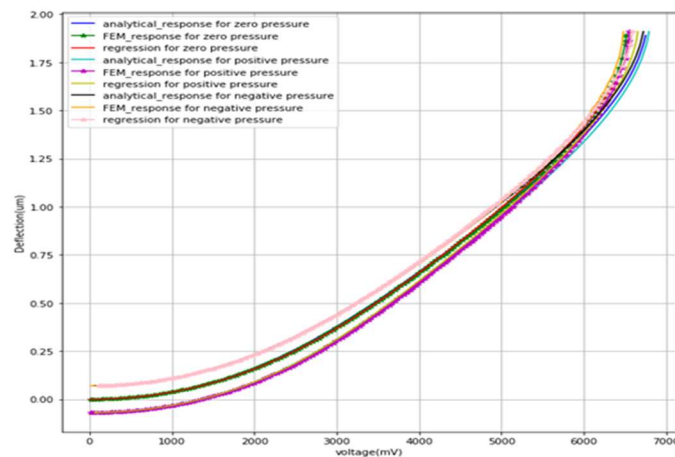


Figure 28 : Voltage vs Deflection plot for prismatic beam with recess

Analytical, simulated, and regression for deflection and voltage at different pressure are compared and plotted in the graph. The deviation in the analytical solution can be noticed because of the limitations of the analytical solution for the prismatic beam with recess. To understand more about the regression model, the graph between voltage and deflection is shown below for zero pressure.

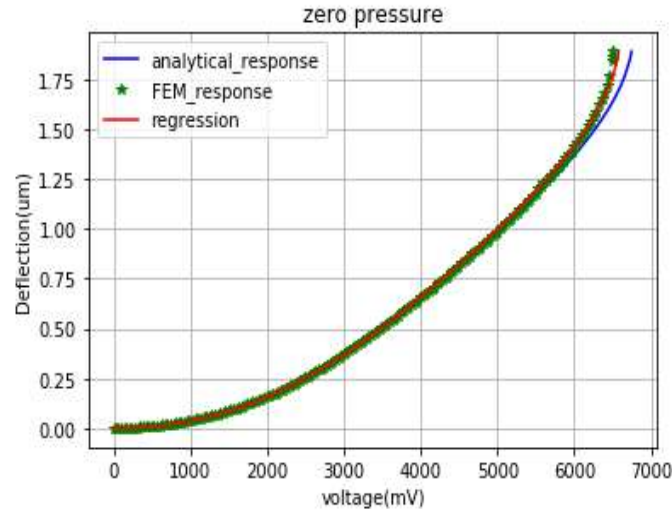


Figure 29 : Voltage vs Deflection plot at zero pressure for prismatic beam with recess

To know the fitness of regression model, errors are calculated and shown below,

Description	units	Analytical model	Regression model	Error in percentage
Spring stiffness k_1	$\mu\text{N}/\mu\text{m}$	0.01548	0.01494	0.05
Spring stiffness k_3	$\mu\text{N}/\mu\text{m}$	0.00985	0.01074	0.08
Effective piston length S_{eff}	$\mu\text{N}/\mu\text{m}/\text{MPa}$	1042.841	1047.245	0.42

Table 12 Calculation of error for the regression model of spring stiffness and effective piston length (prismatic beam with recess)

2. Deflection vs Displaced Area

Same as Case 1, a linear regression model has been implemented for the deflection vs displaced area plot. Equation of linear regression for deflection vs displaced area is,

$$dw = m_1 * dA \quad (33)$$

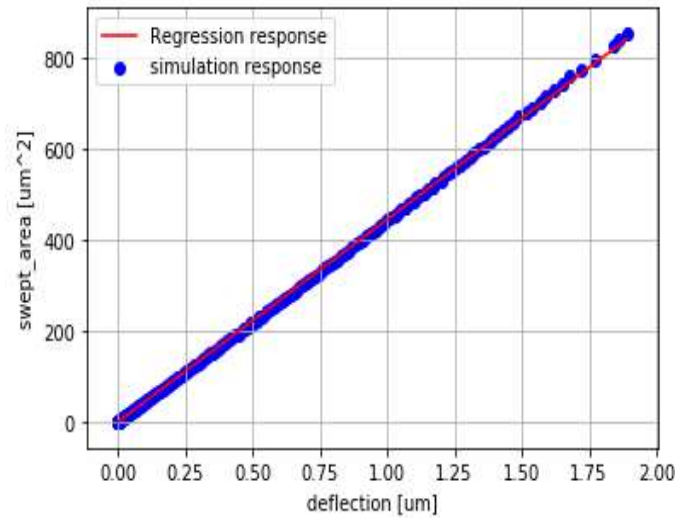


Figure 30 : Deflection vs Displaced Area for prismatic beam with recess

The above graph shows the plot between the regression model and analytical model of the displaced area and deflection. From the graph, it can be observed that the regression solution perfectly fits with the analytical solution.

Equation of regression is as follows,

$$dw = 444.837 * dA \quad (34)$$

Error between displaced area of regression and analytical are as follows:

Description	units	Analytical model	Regression model	error
<i>Displaced area</i>	μm	418.4	444.837	6.31 %

Table 13 Calculation of error for the regression model of displaced area (prismatic beam with recess)

Root mean square error for other set of parameters between displaced area of regression and analytical are as follows:

Description	units	Root mean square error
Displaced area	μm	2.0 %

Table 14 Root mean square of displaced area (prismatic beam with recess)

3. Voltage vs Charge

The charge of the prismatic beam with a recess in each sub-step voltage is calculated in the FEM simulation and stored in the .csv file. Same like the previous study, the regression

model of the approximate function of Taylor series of an odd powers is used for calculation of charge from voltage.

$$Q = f_1 U + f_3 U^3 + f_5 U^5 \quad (35)$$

From the above equation, f_1 is the capacitance at zero voltage. In addition, analytical equation for capacitance at zero voltage is as follows

$$f_{1_{analytical}} \approx \frac{\epsilon_0 L}{g_0} (1 - 0.5727\sigma + 0.3835\sigma^2) \quad (36)$$

Description	units	Analytical	calculated	Error in percentage
Capacitance C_1	$\mu F / \mu m$	2.8136E-09	2.7675E-09	1.63 %

Table 15 Calculation of error for the regression model of Capacitance C_1 (prismatic beam with recess)

Error in the capacitance at zero voltage is 1.6 percent. This shows the fitness of the chosen regression model. Additionally, the root mean square is calculated and shown below.

Description	units	Root mean square error
Charge Q	mV	2.16 %

Table 16 Root mean square of displaced area (prismatic beam with recess)

The root mean square of charge is 2 percent. This shows the charge calculated from the regression model is approximately equal to the analytical solution.

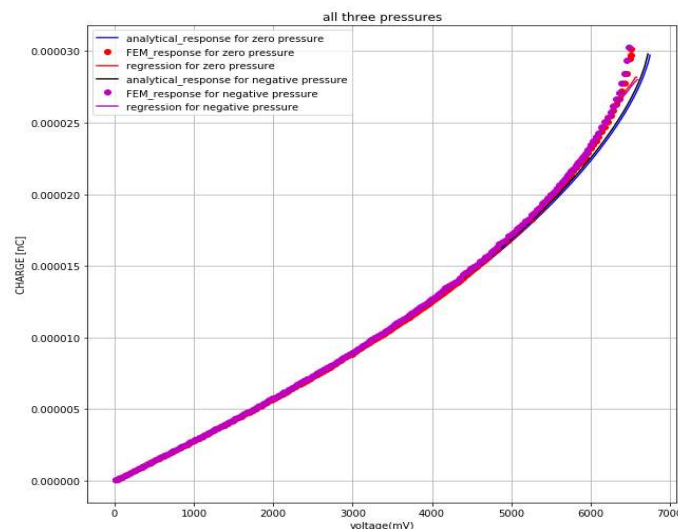


Figure 31 : Voltage vs Charge plot for prismatic beam with recess

Analytical, simulated, and regression for charge and voltage at different pressure are compared and plotted in the graph. Because of the limitations of the analytical solution for the prismatic beam with recess, deviation in the analytical solution can be noticed.

4. Voltage vs Electro Mechanical Resonance Frequency

As in the previous study, the Electromechanical Resonance Frequency of the prismatic beam in each sub-step voltage is calculated in the FEM simulation and stored in the .csv file.

Comparison of simulation, analytical and regression of voltage and electromechanical resonance frequency at zero pressure of simulation is follows,

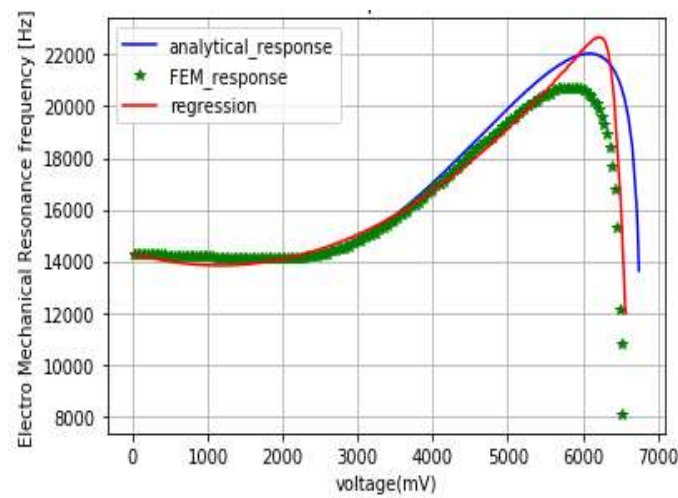


Figure 32 : Voltage vs electromechanical resonance frequency (recess)

A deviation can be noticed in the regression model and, analytical model at the pull-in-region because the regression function fails to have the exact form of the analytical curve, which has few limitations for the prismatic beam with recess. As discussed in the previous case study, the pull-in region is less area of interest from the application's point of view. Only 70 percent of data has been chosen for the regression analysis to reduce the complexity of the analysis with less order equation.

Description	units	Analytical	calculated	Error in percentage
Resonance frequency f_0	Hz	14259.0297	14300	0.28 %

Table 17 Calculation of error for the regression model of Resonance frequency f_0 (prismatic beam with recess)

Error in the capacitance at zero voltage is 0.28%. This shows the fitness of the chosen regression model.

Description	units	Root mean square error	RMSE (90% data)
Electro mechanical resonance frequency	<i>Hz</i>	5.3245	2.79 %

Table 18 Root mean square of electromechanical resonance frequency (prismatic beam with recess)

Compared to 100 percent data, the root-mean-square of 90 percent data gives the error of 2.80%. The system will not operate at the pull-in voltage because it collapses the entire system. Hence, this regression model can be used for the calculation of electromechanical resonance frequency until pull-in voltage.

3.5 GLOBAL ANALYSIS OF PRISMATIC BEAM

Global analysis is the analysis that gives information about how the performance variables depend on geometrical parameters. Global analysis is a regression analysis of an equation with geometrical parameters to predict the performance variables. For regression analysis, dimensional analysis has to be done for the performance variables to make them dimensionless. This dimensional analysis reduces the number of variables and is free from dimensions no matter what scales and units of measurement are selected.

FEM simulation is done for a different set of parameters including their subs step information as mentioned in the results section. A huge database of all geometrical parameters and their performance variables is stored .csv file. A python script with pandas data frame is created for accessing each performance variable according to their geometrical parameter. All these performance variables are converted into dimensionless variables and regression analysis is applied over it. Regression analysis gives an equation with geometrical parameters to predict the performance variables.

1. Resonance frequency

The resonance frequency is the natural frequency of a system at oscillation. To avoid structural or functional issues of the prismatic beam by excessive vibrations, resonance frequencies have to be calculated. By using a modal analysis in FEM simulation, the natural frequency of a prismatic beam under static conditions is calculated for each set of parameters.

Calculated resonance frequencies of a different set of parameters are converted into dimensionless variables for regression analysis.

Dimensionless Resonance frequency of prismatic beam is,

$$\omega'_0 = 2\pi f_0 L \sqrt{\frac{\rho}{E}} \quad (37)$$

Where

f_0 - Mechanical resonance frequency from FEM simulation

L - Length of the prismatic beam

ρ - Density

E – Young's modulus

Resonance frequency depends on the thickness of the prismatic beam. Thick prismatic beam vibrates lower than the thin prismatic beam. Hence, a linear regression for dimensionless thickness b' and dimensionless resonance frequency ω'_0 is carried out and plotted in the graph below.

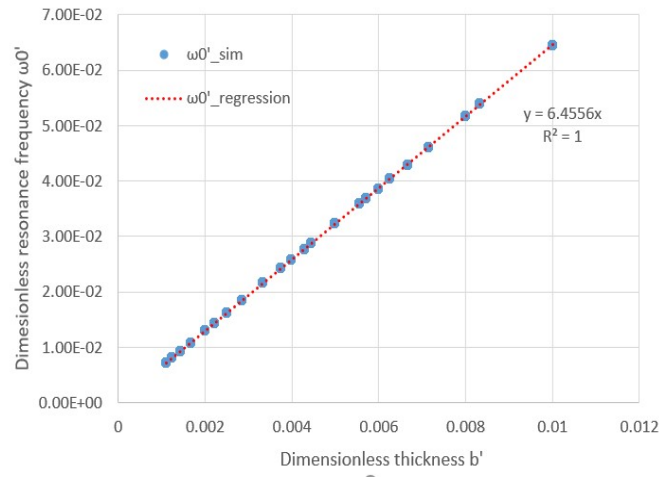


Figure 33 : Linear regression for dimensionless thickness and dimensionless resonance frequency

Root mean square is shown below.

Description	RMSE Error
Mechanical resonance frequency	0.17%

Table 19 Root mean square of mechanical resonance frequency (global analysis)

The above tabulation, it shows the linear regression perfectly fits with the simulated values. Furthermore, constant from analytical solution [15] for finding the resonance frequency is verified with the constant obtained from the linear regression.

Constant from linear regression	Constant ($\beta^2/\sqrt{12}$)	Error in percentage
6.4556	6.458499	0.04%

Table 20 Comparison of constant terms of f_0 - regression and analytical solution (global analysis)

Equation of linear regression model for dimensionless resonance frequency is

$$\omega'_0 = 6.4556 b' \quad (38)$$

With this equation, the dimensionless resonance frequency can be calculated without FEM simulation. By rearranging the above equations, the mechanical resonance frequency of the prismatic beam can be calculated.

2. Spring stiffness k_1

Spring stiffness k_1 is an important parameter to analyse. From local analysis, spring stiffness k_1 of each parametric set is converted into dimensionless variable for regression analysis.

Dimensionless Spring stiffness k'_1 is,

$$k'_1 = \frac{k_1}{E} \quad (39)$$

Where

k_1 – Spring stiffness form local analysis of prismatic beam.

E – Young's modulus

Spring stiffness depends on the thickness of the prismatic beam. Hence, a polynomial regression of degree 3 is used as regression analysis for dimensionless thickness b' and dimensionless spring stiffness k_1 is carried out and plotted in the graph below.

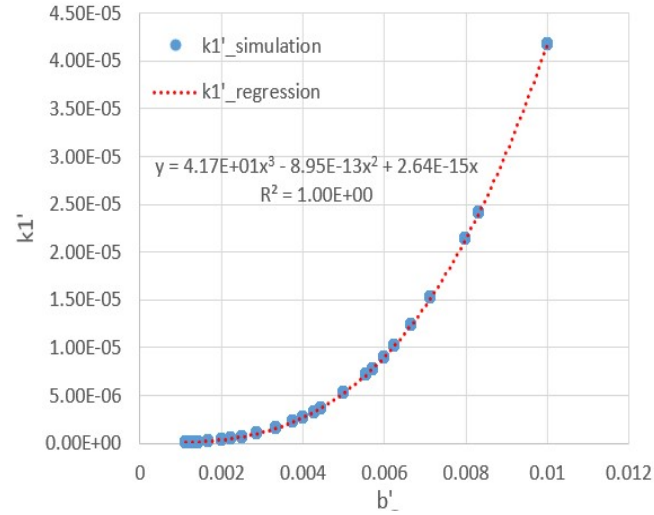


Figure 34 : Polynomial regression for dimensionless thickness and dimensionless spring stiffness

Root mean square is shown below.

Description	RMSE Error
Spring stiffness K1	0.09%

Table 21 Root mean square of spring stiffness k_1 (global analysis)

The above tabulation, it shows the polynomial regression perfectly fits with the simulated values. Furthermore, constant from analytical solution for finding the spring stiffness is verified with the constant obtained from the polynomial regression.

Constant from linear regression	Constant ($\frac{\lambda_0}{12}$)	Error in percentage
41.7	41.7136	0.03%

Table 22 Comparison of constant terms of k_1 - regression and analytical solution (global analysis)

Equation of polynomial regression model for dimensionless spring stiffness is

$$k'_1 = 41.7 * b'^3 \quad (40)$$

With this equation, dimensionless spring stiffness can be calculated without FEM simulation. By rearranging the above equations, the spring stiffness of the prismatic beam can be calculated.

3. Capacitance C_1

The capacitance of the prismatic beam gives the amount of charge that it can hold. With higher capacitance, the higher charge can prismatic beam hold. Therefore, Capacitance C_1 is an important parameter to analyse. From local analysis, the Capacitance C_1 of each parametric set is converted into a dimensionless variable for regression analysis.

Dimensionless Capacitance c'_1 is,

$$c'_1 = \frac{C_1}{\epsilon_0 \epsilon_r} \quad (41)$$

Where

C_1 - Capacitance form local analysis of prismatic beam.

$\epsilon_0 \epsilon_r$ - Permittivity

$\sigma = \left(\frac{s}{g_0} \right)$, where s is the recess and g_0 is electrostatic gap

Capacitance depends on the recess and electrostatic gap of the prismatic beam. Hence, a polynomial regression of degree 2 is used as regression analysis for sigma and dimensionless Capacitance c'_1 times dimensionless electrostatic gap is carried out and plotted in the graph below.

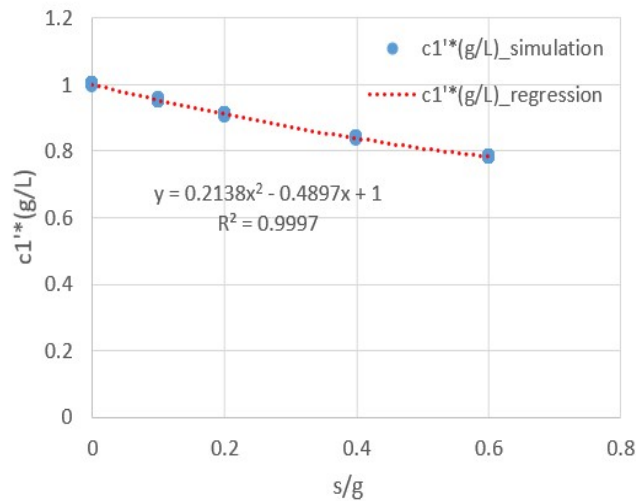


Figure 35 : Polynomial regression for Capacitance C_1

Root mean square is shown below.

Description	RMSE Error
Regression Capacitance c'_1	0.16%

Table 23 Root mean square of Capacitance C1 (global analysis)

From the above tabulation, it shows the polynomial regression perfectly fits with the simulated values. Equation of polynomial regression model for dimensionless capacitance c'_1 is

$$c'_1 * g' = 0.2138 \left(\frac{s}{g}\right)^2 - 0.4897 \left(\frac{s}{g}\right) + 1 \quad (42)$$

With this equation, dimensionless capacitance can be calculated without FEM simulation. By rearranging the above equations, the dimensionless capacitance of the prismatic beam can be calculated.

4. Pull-in-voltage U_{PI}

Pull-in of a beam is an unstable equilibrium state and critical voltage is called pull-in-voltage. Beyond the pull-in-voltage, the beam collapse completely. Therefore, Pull-in-voltage U_{PI} is an important parameter to analyze. By using static analysis with the input voltage in FEM simulation, the pull-in-voltage of a prismatic beam under static conditions is calculated for each set of parameters. Calculated pull-in-voltage of a different set of parameters are converted into dimensionless variables for regression analysis.

Dimensionless Pull in voltage u'_{PI}

$$u'_{PI} = \frac{U_{PI}}{L} \sqrt{\frac{\epsilon_0 \epsilon_r}{E}} \quad (43)$$

Where,

ϵ_0 - Relative permittivity

ϵ_r - Vacuum permittivity

E - Young's modulus

ρ - Density

L - Length

Pull-in-voltage depends on the thickness of the prismatic beam. Thick prismatic beam have higher pull-in-voltage than the thin prismatic beam. Hence, a polynomial regression for

dimensionless thickness b' and dimensionless pull-in-voltage u'_{PI} times dimensionless electrostatic gap of power 1.5 is carried out and plotted in the graph below.

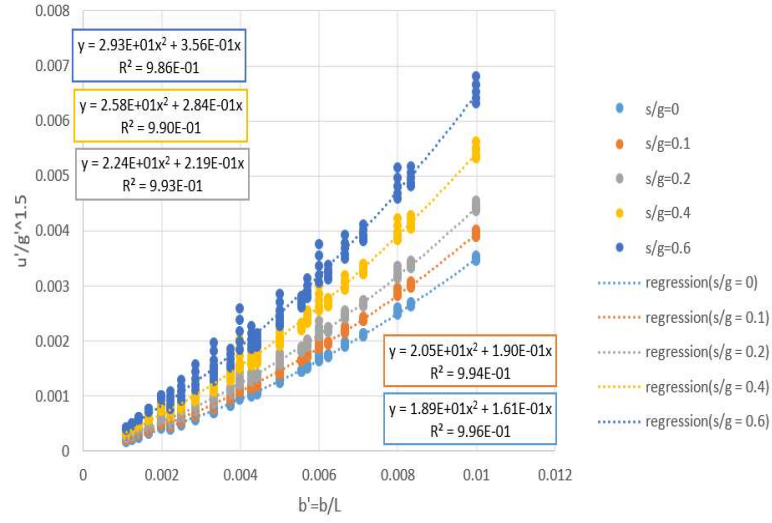


Figure 36 : Polynomial regression for Pull-in-voltage (global analysis)

Polynomial regression of degree 2 is used for different sigma values (i.e. $\sigma = \left(\frac{s}{g_0}\right)$). To generalize the polynomial equation, different constants for different sigma values are calculated and separate regression model created.

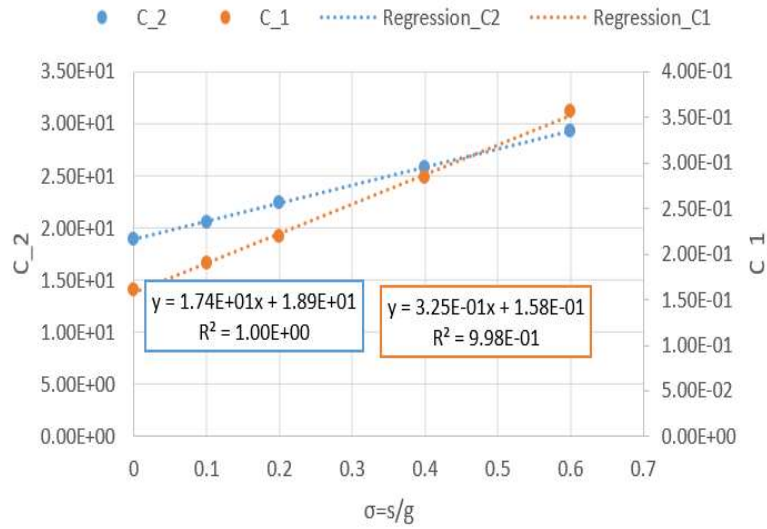


Figure 37 : Polynomial regression for constants of Pull-in-voltage (global analysis)

Therefore, general equation of regression model of Dimensionless Pull in voltage u'_{PI} is

$$\frac{u'}{g'^{1.5}} = (0.158 + 0.325\sigma)b' + (18.9 + 17.4\sigma)b'^2 \quad (44)$$

With this equation, dimensionless pull-in-voltage can be calculated without FEM simulation. By rearranging the above equations, the pull-in-voltage of the prismatic beam can be calculated.

Root mean square is shown below.

Description	RMSE Error
Regression Dimensionless Pull in voltage u'_{PI}	9.7%

Table 24 Root mean square of Capacitance C1 (global analysis)

Root mean square is an overall error of the pull-in-voltage for a different set of parameters. From the analysis, it is understood that for higher recess, the regression model fails to calculate the exact pull-in-voltage. However, it is a good approximation for the prismatic beam with less recess.

3.6 S-SHAPED NED BEAM – COMPARATIVE STUDY

Same like prismatic beam, local analysis is carried out for finding the information of how the performance variables depends on input voltage.

Parametric study of different geometrical parameters has been done for the S-shaped NED beam. Only two set of the parameters is taken into account to show the results.

1. S-shaped NED beam with electrostatic gap = 1
2. S-shaped NED beam with electrostatic gap = 2.5

3.6.1 CASE 1: S-shaped NED beam with electrostatic gap = 1

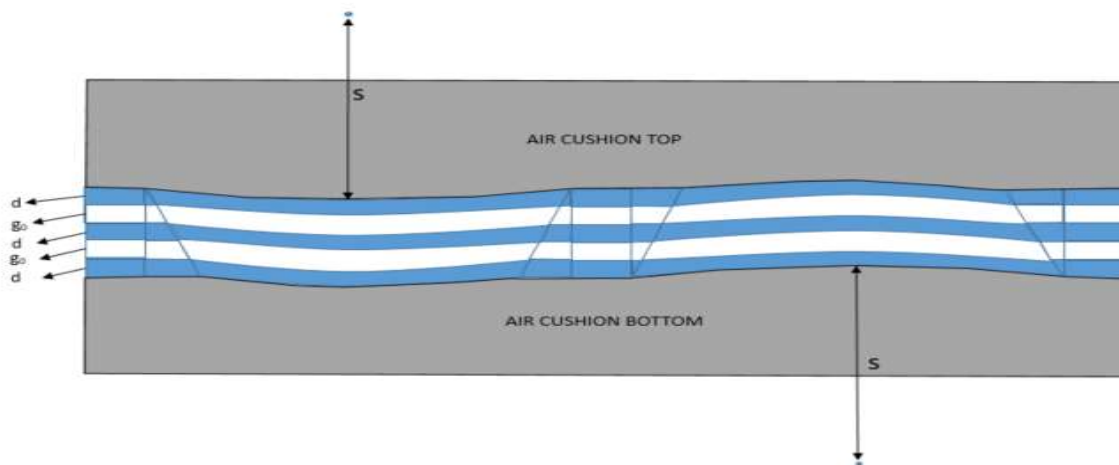


Figure 38 : S-shaped NED beam of gap $g_0 = 1$

Parameters considered are

Arc Length = $700\ \mu m$
Thickness = $1\ \mu m$
Electrostatic Gap = $1\ \mu m$
Alpha = 10 degree

When voltage is applied in the electrostatic gap, an electrostatic force is created between the top and bottom electrodes. These electrostatic electrodes, in turn, transform electrostatic forces into mechanical forces. Therefore, top and bottom electrodes tend to bend towards the middle electrode and cause the deflection at end of the s-shaped beam. Same as the prismatic beam, FEM simulation of static, modal, and linear perturbation analysis is done in Ansys APDL and results are as follows.

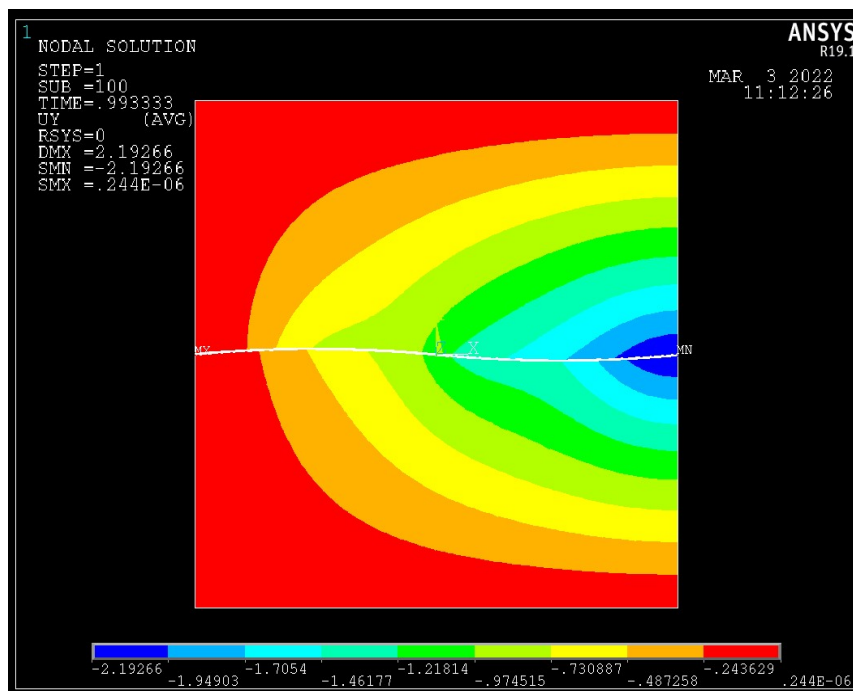


Figure 39 : FEM results of S shaped beam of gap $g_0 = 1$

Post processing:

1. Voltage Vs Deflection

Same as the prismatic beam, the following equation is used in the Powell's method for the regression model for spring stiffness.

$$k_1 x_{electrode} + k_2 x_{electrode}^2 + k_3 x_{electrode}^3 = \frac{1}{2} \frac{C_0}{g_0} U_{in}^2 f'(w) \quad (45)$$

To retrieve the effective electrode displacement from nodal deflection,

$$x_{electrode} = w/\theta \quad (46)$$

Where,

k_1 – Spring stiffness 1

k_2 – Spring stiffness 2

k_3 – Spring stiffness 3

θ – Theta for retrieve the effective electrode displacement

w – Deflection at pull-in

C_0 – Capacitance at 0V

g_0 – Electrostatic gap

U_{in} – Input Voltage

$$f'(w) = \frac{1}{(g_0 - x_{electrode})^2}$$

DESCRIPTION	UNITS	REGRESSION VALUES
spring stiffness 1 (k_1)	$\mu N/\mu m^2$	1.7079
spring stiffness 2 (k_2)	$\mu N/\mu m^2$	1.9071
spring stiffness 3 (k_3)	$\mu N/\mu m^2$	4.1936
Theta (θ)	<i>radians</i>	8.4827

Table 25 Calculated spring stiffness and effective piston length from regression model for S-shape beam ($g_0 = 1$)

With the above co-efficient values, the deflection for any voltage can be calculated. Plot for FEM simulation and regression for voltage vs deflection is shown below.

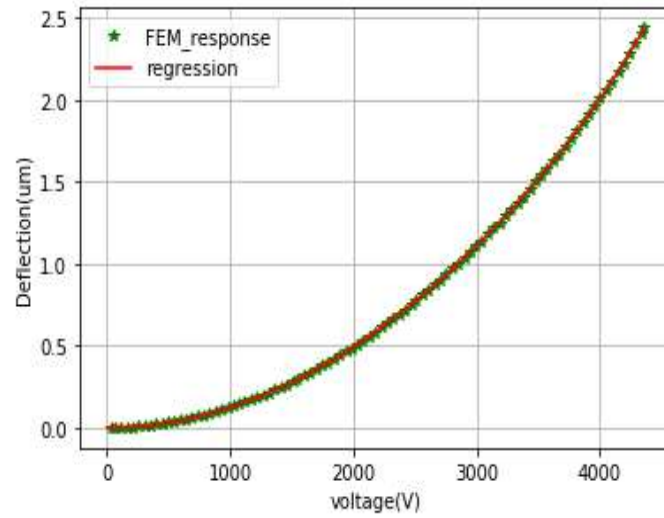


Figure 40 : Voltage vs Deflection plot for S-shaped beam of gap $g_0 = 1$

The above graph shows that the regression model perfectly fits with the simulation analysis. Hence, the deflection for any voltage can be calculated from the regression model.

2. Deflection vs Displaced Area

Since two air cushions are considered on top and bottom of the electrodes, two displaced area has to be calculated for the top and bottom of the electrodes respectively.

When voltage is applied, the top air cushion expands, and the bottom air cushion contracts. The basic assumption of the analytical model is that only the first mode of oscillation is active, and therefore the displaced area is directly proportional to the deflection. Hence, linear regression model of $y = mx + C$ is used. The displaced area at zero voltage does not make any use for the calculation hence the constant term is forcefully made into zero. Therefore, the equation of linear regression for top air cushion is,

$$dw = m_1 * dA \quad (47)$$

From linear regression model, slope $m_1 = 1462.859$ and r^2 value is 0.999

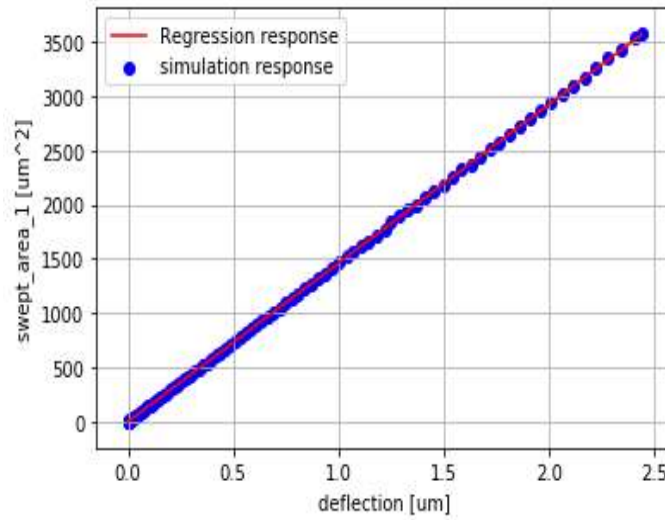


Figure 41 : Deflection vs Displaced Area Top cushion for S-shaped beam of gap $g_0 = 1$

Equation of linear regression for bottom air cushion is,

$$dw = m_1 * dA \quad (48)$$

From linear regression model, slope $m_1 = -1342.642$ and r^2 value is 0.999

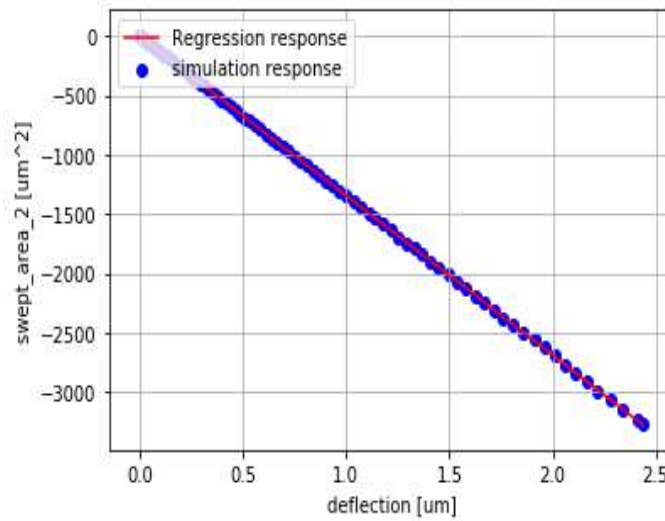


Figure 42 : Deflection vs Displaced Area bottom cushion for S-shaped beam of gap $g_0 = 1$

r^2 Value of 0.99 shows the regression model perfectly fits with the simulation values.

3. Voltage vs Charge

Charge of the S shaped beam in each sub step voltage is calculated in the FEM simulation and stored in the .csv file

An approximate function is created with the odd powers of Taylor series for regression model. Equation of odd powers of Taylor series regression for charge is,

$$Q = C_1 U + C_3 U^3 \quad (49)$$

From the regression model, Constants $C_1 = 4.9377\text{e-}08$, $C_2 = 1.6310\text{e-}16$

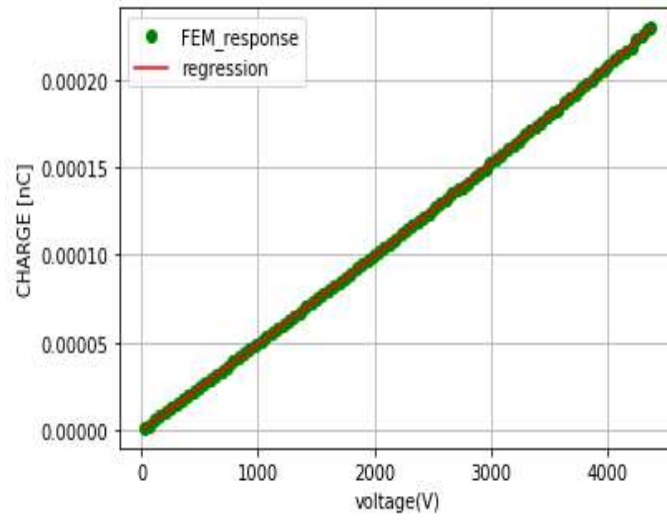


Figure 43 : Voltage vs Charge plot for S-shaped beam of gap $g_0 = 1$

The above graph shows that the regression model perfectly fits with the simulation analysis. Hence, the charge at any voltage can be calculated from the regression model.

4. Voltage vs Electromechanical Resonance Frequency

From the linear perturbation analysis, the Electromechanical Resonance Frequency of the S-shaped beam in each sub-step voltage is calculated in the FEM simulation and stored in the .csv file.

An approximate function is created with the Taylor series and the sigmoid function for the regression model. Equation of regression model for Electromechanical Resonance Frequency is,

$$f_{EM} = f_0 * \left(\frac{1}{1 + e^{\frac{U - U_{pi}}{\beta}}} \right) * (1 + (f_1 U + f_2 U^2)) \quad (50)$$

From the regression model, Constants $f_0 = 1920$, $f_1 = -5.9017\text{e-}06$, $f_2 = 1.2468\text{e-}08$ and $\beta = \frac{U_{pi}}{200}$.

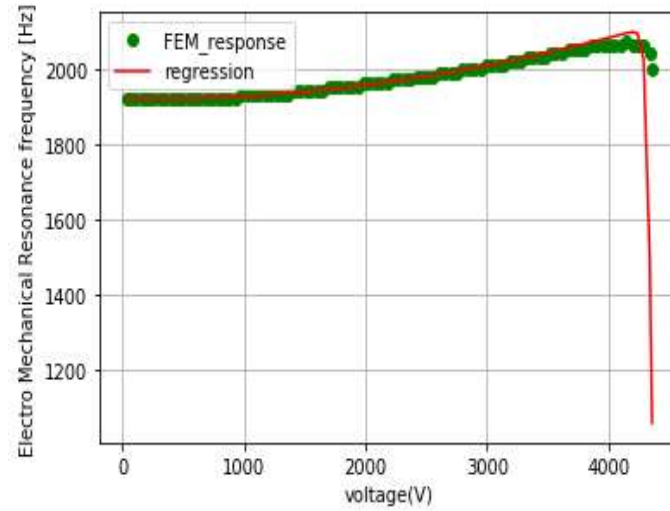


Figure 44 : Electromechanical resonance frequency vs Charge plot, gap $g_0 = 1$

A deviation can be noticed in the regression model at the pull-in-region because the function fails to have the exact form of the simulation curve. From an application point of view, the system cannot operate at the pull-in region. Therefore, it becomes less area of interest. Weighted regression is used in this analysis. 70 percent of data has been chosen for the regression analysis to reduce the complexity of the analysis with less order equation.

3.6.2 CASE 2: S-shaped NED beam with electrostatic gap = 2.5

Parameters considered are

Arc Length = $700 \mu m$
Thickness = $1 \mu m$
Electrostatic Gap = $2.5 \mu m$
Alpha = 10 degree

Same as the previous case study, FEM simulation of static, modal and linear perturbation analysis are done in Ansys APDL and results are as follows.

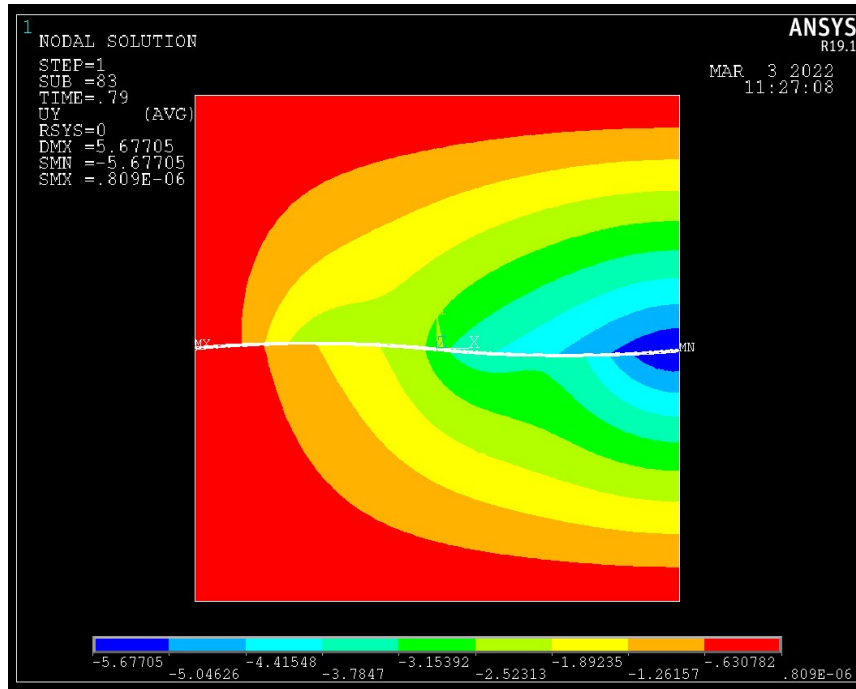


Figure 45 : FEM results of S shaped beam of gap $g_0 = 2.5$

Post processing:

1. Voltage vs Deflection

Same as the previous case study, Powell's method is used for calculation of spring stiffness.

DESCRIPTION	UNITS	REGRESSION VALUES
spring stiffness 1 (k_1)	$\mu\text{N}/\mu\text{m}^2$	1.1127
spring stiffness 2 (k_2)	$\mu\text{N}/\mu\text{m}^2$	1.9071
spring stiffness 3 (k_3)	$\mu\text{N}/\mu\text{m}^2$	3.3943
Theta (θ)	radians	9.4958

Table 26 Calculated spring stiffness and effective piston length from regression model for S-shape beam ($g_0 = 2.5$)

Plot for FEM simulation and regression for voltage vs deflection is shown below.

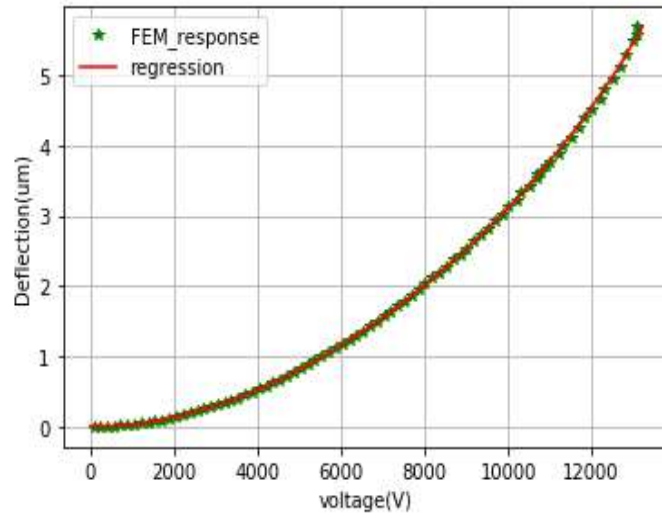


Figure 46 : Voltage vs Deflection plot for S-shaped beam of gap $g_0 = 2.5$

The above graph shows that the regression model perfectly fits with the simulation analysis. Hence, the deflection for any voltage can be calculated from the regression model.

2. Deflection vs Displaced Area

Same as the previous case study, linear regression is used for the calculation of the displaced area. Equation of linear regression for top air cushion is,

$$dw = m_1 * dA \quad (51)$$

From linear regression model, slope $m_1 = 1510.111$ and r^2 value is 0.999

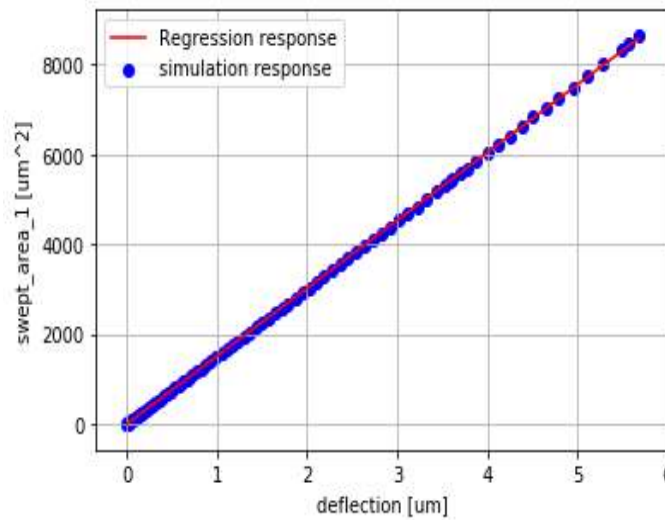


Figure 47 : Deflection vs Displaced Area Top cushion for S-shaped beam of gap $g_0 = 2.5$

Equation of linear regression for bottom air cushion is,

$$dw = m_1 * dA \quad (52)$$

From linear regression model, slope $m_1 = -1294.645$ and r^2 value is 0.999

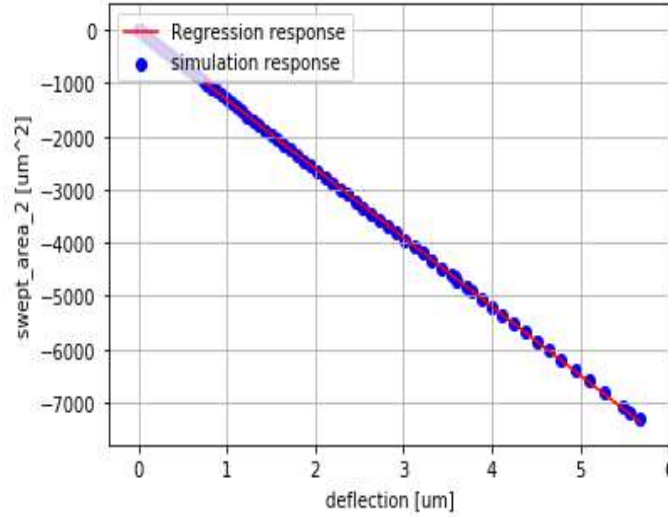


Figure 48 : Deflection vs Displaced Area bottom cushion for S-shaped beam of gap $g_0 = 2.5$

r^2 Value of 0.99 shows the regression model perfectly fits with the simulation values.

3. Voltage vs Charge

Same as the previous case study, odd powers of Taylor series for regression model is used for calculation of Charge.

$$Q = C_1 U + C_3 U^3 \quad (53)$$

From the regression model, Constants $C_1 = 1.9576e-08$, $C_2 = 1.3093e-17$

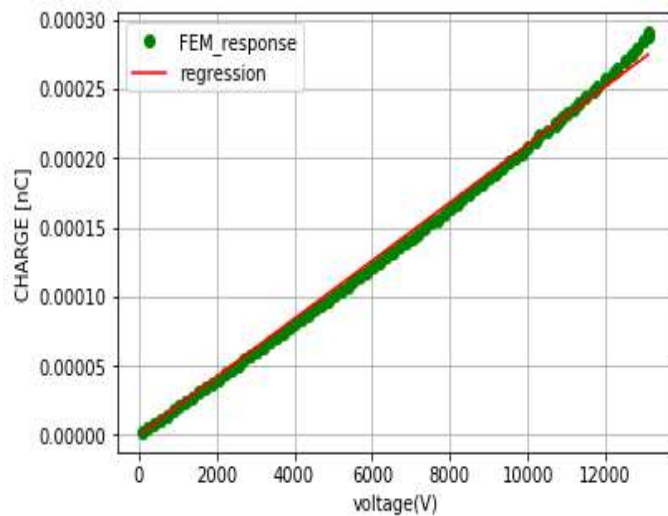


Figure 49 : Voltage vs Charge plot for S-shaped beam of gap $g_0 = 2.5$

The above graph shows that the regression model perfectly fits with the simulation analysis. Hence, the charge at any voltage can be calculated from the regression model.

4. Voltage vs Electro Mechanical Resonance Frequency

Same as the previous case study, Taylor series and the sigmoid function as regression model is used for calculation of Charge.

$$f_{EM} = f_0 * \left(\frac{1}{1 + e^{\frac{U - U_{pi}}{\beta}}} \right) * (1 + (f_1 U + f_2 U^2)) \quad (54)$$

From the regression model, Constants $f_0 = 1920$, $f_1 = -5.2636e-07$, $f_2 = 3.0272e-09$ and $\beta = \frac{U_{pi}}{200}$.

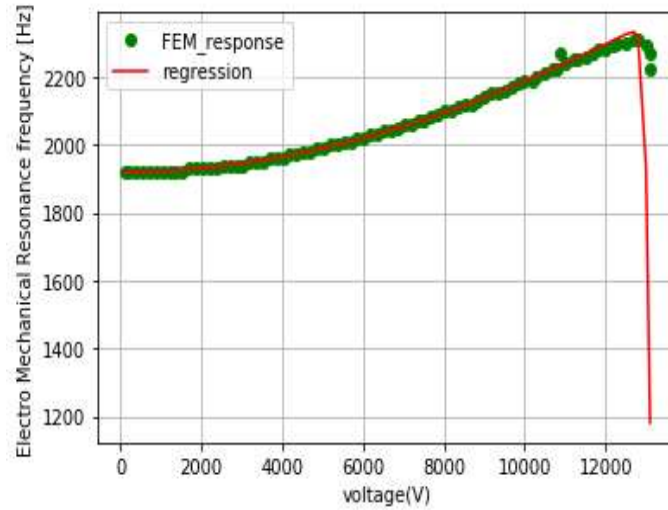


Figure 50 : Electromechanical resonance frequency vs Charge plot, gap $g_0 = 2.5$

A deviation can be noticed in the regression model at the pull-in-region because the function fails to have the exact form of the simulation curve. From an application point of view, the system cannot operate at the pull-in region. Therefore, it becomes less area of interest. Weighted regression is used in this analysis. 70 percent of data has been chosen for the regression analysis to reduce the complexity of the analysis with less order equation.

3.7 GLOBAL ANALYSIS OF S-SHAPED NED BEAM

Same as a prismatic beam, global analysis for an S-shaped beam has been done to predict the performance variables.

FEM simulation is done for a different set of parameters including their subs step information as mentioned in the results section. A huge database of all geometrical parameters and their performance variables is stored .csv file. A python script with pandas data frame is

created for accessing each performance variable according to their geometrical parameter. All these performance variables are converted into dimensionless variables and regression analysis is applied over it. Regression analysis gives an equation with geometrical parameters to predict the performance variables.

1. Resonance frequency

As discussed in the global analysis of prismatic beam, the resonance frequency is calculated from modal analysis in FEM simulation. Resonance frequency gives information about the structural and functional issues of the S-shaped beam. Calculated resonance frequencies of a different set of parameters are converted into dimensionless variables for regression analysis. The dimensionless Resonance frequency of prismatic beam is,

$$\omega'_0 = 2\pi f_0 L \sqrt{\frac{\rho}{E}} \quad (55)$$

Where

f_0 - Mechanical resonance frequency from FEM simulation

L - Length of the prismatic beam

ρ - Density

E – Young's modulus

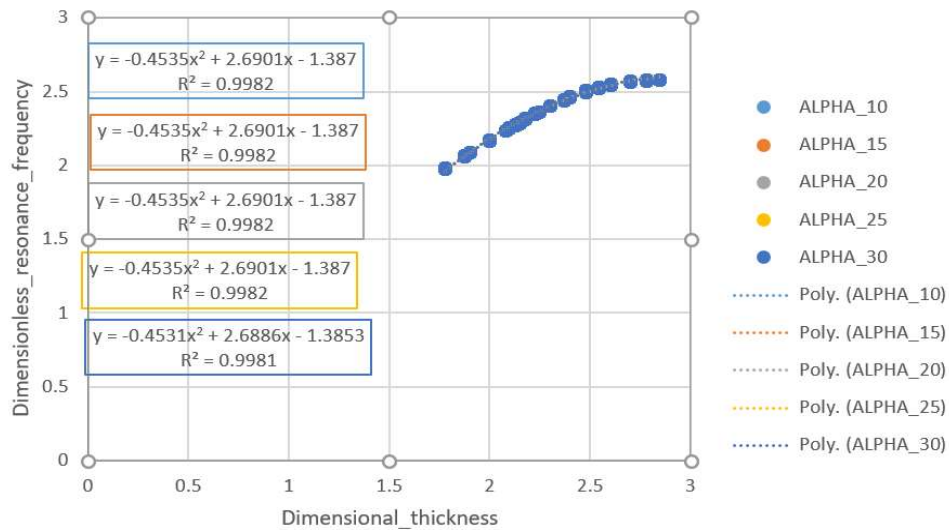


Figure 51 : Linear regression of dimensionless thickness and dimensionless resonance frequency for S-shaped beam

Root mean square is shown below.

Description	RMSE Error
Mechanical resonance frequency	0.37%

Table 27 Root mean square of mechanical resonance frequency S-shaped beam (global analysis)

From the above tabulation, it shows the linear regression perfectly fits with the simulated values. Equation of linear regression model for dimensionless resonance frequency is

$$\log(\omega'_0) = (-0.4535 * \log(b'^2)) + (2.6901 * \log b') - 1.3887 \quad (56)$$

With this equation, dimensionless resonance frequency can be calculated without FEM simulation. By rearranging the above equations, mechanical resonance frequency of the S-shaped beam can be calculated.

CHAPTER 4

SUMMARY AND OUTLOOK

This research aimed to find the empirical laws that relate the performance variables with the geometrical parameters. Nonlinear NED behavior such as Static and modal analysis of Prismatic coulomb actuated beam is done successfully by simulating the structure in the finite element method. The parametric sweep of prismatic beam for clamped-clamped structure is performed using finite element simulations. The local analysis is carried out for the performance variables (pull-in voltage, deflection, mechanical resonance frequency, charge, displaced area, and electromechanical resonance frequency) of each parametric set (length, thickness, electrostatic gap and recess) of the prismatic beam to find the dependency on these variables on the input voltage. A regression model for each performance variable is identified to fit with the simulation results. Furthermore, the results from the regression model are verified with a known analytical solution to prismatic beam [15]. Thus, the proposed regression model of the local analysis can relate the performance variables with input voltage.

In addition, the global analysis is performed for all the performance variables of each parametric set obtained from the local analysis. Performance variables depending on the dimensions of the geometrical parameters can cause the model to be inhomogeneity and dimensionally dependent. Moreover, to reduce the number of variables for the regression model, dimensional analysis of Buckingham's Pi theorem is applied to performance variables and geometrical parameters. As a result of global analysis, an empirical relationship of the prismatic beam between its geometrical parameters and its electromechanical performance is identified. With the outcome of regression analysis, each performance variable can be calculated without simulating the NED structure. Thus, the proposed regression model of each performance variable reduces the computational time and cost.

This entire procedure is repeated for the complex structure of the S-shaped NED beam. The local analysis of the S-shaped NED beam shows similar results as compared to the local analysis of the prismatic beam. Thus, the performance variables can be calculated for the corresponding input voltage of the S-shaped NED beam. As an extension of the current work, we could perform a global analysis on the S-shaped NED beam and validate its behavior as per expectation concerning the prismatic beam.

ANNEXURE - I

PROOF FOR DIMENSIONAL ANALYSIS OF PERFORMANCE VARIABLES

1. RESONANCE FREQUENCY

$$\text{Dimensionless Resonance Frequency} = \omega_n S \sqrt{\frac{\rho}{E}}$$

Units

$$\text{Resonance frequency } (\omega_n) = T^{-1}$$

$$\text{Length (S)} = L$$

$$\text{Density } (\rho) = ML^{-3}$$

$$\text{Young's modulus } = E = ML^{-1}T^{-2}$$

$$\text{Proof: } \omega_n S \sqrt{\frac{\rho}{E}} = 1$$

$$\text{LHS} = T^{-1} * L * \sqrt{\frac{ML^{-3}}{ML^{-1}T^{-2}}}$$

$$= T^{-1} * L * \sqrt{\frac{L^{-2}}{T^{-2}}}$$

$$= 1$$

$$= \text{RHS}$$

Hence proved

2. PULL IN VOLTAGE

$$\text{Dimensionless Pull in voltage} = \frac{U}{S} \sqrt{\frac{\epsilon_0 \epsilon_r}{E}}$$

Units

$$\text{Voltage } (U) = ML^2T^{-3}A^{-1}$$

$$\text{Absolute Permittivity } (\epsilon_0 \epsilon_r) = M^{-1}L^{-3}A^2T^4$$

$$\text{Young's modulus } (E) = ML^{-1}T^{-2}$$

$$\text{Length (S)} = L$$

$$\text{Proof: } \frac{U}{S} \sqrt{\frac{\epsilon_0 \epsilon_r}{E}} = 1$$

$$\begin{aligned}
\text{LHS} &= \frac{ML^2T^{-3}A^{-1}}{L} * \sqrt{\frac{M^{-1}L^{-3}T^4A^2}{ML^{-1}T^{-2}}} \\
&= \frac{ML^2T^{-3}A^{-1}}{L} * \sqrt{M^{-2}L^{-2}T^4A^2T^2} \\
&= MLT^{-3}A^{-1}M^{-1}L^{-1}T^2AT \\
&= 1 \\
&= \text{RHS}
\end{aligned}$$

Hence proved

3. CHARGE

$$\text{Dimensionless charge} = \frac{Q}{S\sqrt{\epsilon_0\epsilon_rE}}$$

Units

$$\text{Charge } Q \text{ (per unit length)} = ATL^{-1}$$

$$\text{Absolute Permittivity } (\epsilon_0\epsilon_r) = M^{-1}L^{-3}A^2T^4$$

$$\text{Young's modulus } (E) = ML^{-1}T^{-2}$$

$$\text{Length (S)} = L$$

$$\textbf{Proof: } \frac{Q}{S\sqrt{\epsilon_0\epsilon_rE}} = 1$$

$$\begin{aligned}
\text{LHS} &= \frac{AT^{-1}L^{-2}}{\sqrt{M^{-1}L^{-3}T^4A^2} \sqrt{ML^{-1}T^{-2}}} \\
&= \frac{AT^{-2}}{\sqrt{L^{-3}L^{-1}T^2A^2}} \\
&= \frac{AT^{-2}}{\sqrt{L^{-4}T^2A^2}} \\
&= \frac{AT^{-2}}{ATL^{-2}} \\
&= 1
\end{aligned}$$

Hence proved

ANNEXURE - II

ANALYTICAL SOLUTION OF COULOMB-ACTUATED MICROBEAM

The differential equation of coulomb actuated prismatic beam is

$$EI \frac{\partial^4 w}{\partial x^4} + \mu \frac{\partial^2 w}{\partial t^2} + \delta \frac{\partial w}{\partial t} = \frac{1}{2} \epsilon_0 \epsilon_r h \frac{U^2(t)}{[g(x) - w(x, t)]^2} - ph + \frac{EA}{2L} \left[\int_{-L/2}^{L/2} \left(\frac{\partial w}{\partial x'} \right)^2 dx' \right] \frac{\partial^2 w}{\partial x^2} \quad (57)$$

By H.A.G. Schenk's solution [15], the above equation can be written as an ordinary differential equation that describes the motion of the midpoint of the beam (w_0).

The corresponding lumped-parameter differential equation is the following:

Force balance

$$m\ddot{w}_0 + r\dot{w}_0 + k_1 w_0 + k_3 w_0^3 = \frac{1}{2} \frac{C_0}{g_0} U_{in}^2 F\left(\frac{w_0}{g_0}\right) - F_{out} \quad (58)$$

Parameters:

- $k_1 = \frac{\lambda_0 EI}{L^3}$
- $k_3 = \frac{\epsilon k_1}{g_0^2}$
 - $\epsilon = 6 \left(\frac{g_0}{b} \right)^2 \delta$
- $m = \mu L$
 - $\omega_0 = \left(\frac{\beta_0}{L} \right)^2 \sqrt{\frac{EI}{\mu}} = \frac{b}{L^2} \frac{\beta_0^2}{\sqrt{12}} \sqrt{\frac{E}{\rho}}$
- $C_0 = \frac{\epsilon_0 \epsilon_r h L}{g_0}$
- $F_{out} = p_{out} h L$

Generally, asymmetric electro-static transducers are modeled as a spring capacitor system. A parallel plate capacitor (C_0) with an initial gap of g_0 applied with a voltage U_{in} across the electrode tends to displace the moving electrode towards the fixed one. Such displacement reduces the gap between the electrodes. However, it is opposed by the force of a polynomial spring k_i . Hence, higher-order terms are introduced to model the stress-stiffening effects. The term F_{out} gives the additional loads to the system.

Electrical and mechanical constitutive equations are

$$q(t) = \frac{C_0}{1 - \left(\frac{w(t)}{g_0}\right)} U_{in}(t) \quad (59)$$

$$k_1 w + k_3 w^3 = \frac{1}{2} \frac{C_0}{g_0} \frac{U_{in}^2}{(g_0 - w)^2} - F_{out} \quad (60)$$

Though the spring capacitor system gives a good approximation for electrostatic transducers, parallel plate capacitor imposes few limitations. Therefore, this transducer modeled with displacement varying capacitance with polynomial expression.

Capacitance of a prismatic beam as follows,

$$q(t) = C_0(1 + f(w))U_{in}(t) \quad (61)$$

$$k_1 w + k_3 w^3 = \frac{1}{2} \frac{C_0}{g_0} U_{in}^2 f(w) - pS_{eff,2}^* \quad (62)$$

Where p is the pressure applied and $S_{eff,2}^*$ is the effective piston length of sensor. $f'(w)$ gives the variation of capacitance against a reference displacement.

Where,

$$f(w) = \psi^2 \left[\pi_1 + \frac{\pi_2}{\left(1 - \frac{(w - s_0)}{g_0}\right)^{\frac{1}{2}}} + \frac{\pi_3}{\left(1 - \frac{(w - s_0)}{g_0}\right)^{\frac{3}{2}}} \right] \quad (63)$$

where π being constants, $\pi_1 = \frac{\psi^2}{77}$, $\pi_2 = -\frac{\psi^2}{38}$, $\pi_3 = \frac{15\psi^2}{28}$ with $\psi = 1.5881$

$$C_0 = \frac{\epsilon_0 \epsilon_r L}{g_0} \quad (64)$$

ϵ_0 - Relative permittivity

ϵ_r - Vacuum permittivity

L - Length of the beam in μm

b - Thickness of beam in μm

w - Deflection in μm

Therefore,

$$U = \left[(k_1^* w + k_3^* w^3 + pS^*) \frac{2g_0}{C_0^* f(w)} \right]^{\frac{1}{2}} \quad (65)$$

This equation is used as main equation for voltage vs deflection regression.

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